University of Cebu-Main Campus
College of Computer Studies

FACECHECK: AN AI-POWERED FACIAL RECOGNITION SYSTEM FOR AUTOMATED ATTENDACE TRACKING

A Proposal
to the Faculty of the
College of Computer Studies, University of Cebu

In Partial Fulfillment of the Requirements
For the Subject CAPSTONE4

By

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APPROVAL SHEET

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This Capstone Project Study titled **FACECHECK: AN AI-POWERED FACIAL RECOGNITION SYSTEM FOR AUTOMATED ATTENDACE TRACKING** prepared and submitted by Jhanna Kris Durano, Sean Joseph Arcana, Niño Rollane Ocliasa, and Rovic Steve Real has been examined and is recommended for approval and acceptance.

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We sincerely express our gratitude to all those who played a part in the development and completion of FaceCheck, our AI-powered facial recognition system for attendance tracking.

Our deepest appreciation goes to our adviser, Engr. Dennis Durano, whose guidance, constructive suggestions, and constant support gave us direction and clarity throughout the project. His encouragement helped us stay focused and confident from start to finish.

We are also thankful to the academic community that fostered our growth and learning. The opportunities and support provided throughout our time in the program allowed us to develop not only the technical skills but also the discipline needed to complete this study.

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We are truly grateful to our families for being our constant source of strength. Their patience, understanding, and motivation gave us the energy to push through even the most difficult stages of this project.

Lastly, we thank our friends and teammates, whose collaboration and shared determination made this journey not just successful, but also inspiring and unforgettable.

DEDICATION

We dedicate this project to the students and faculty of our university, whose hard work and passion for learning inspired us to create FaceCheck. Their daily efforts reminded us of the real value of education and the importance of building tools that can make a difference.

To our families, thank you for being our constant support system. Your patience, love, and encouragement helped us stay strong through the many ups and downs of this journey. This achievement is just as much yours as it is ours.

We also dedicate this work to the mentors who guided us along the way. Your words of advice and your belief in our potential have shaped the way we think, grow, and face challenges.

To those who will come after us—future students, innovators, and dreamers—may this project be a stepping stone for even greater ideas. We hope it encourages you to keep pushing boundaries and improving the world through technology.

Above all, this is for the continuous effort to grow, to improve, and to build a future where technology empowers people and brings positive change to learning environments everywhere.

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CHAPTER I

INTRODUCTION

Rationale of the Study

Attendance monitoring constitutes a fundamental component of academic administration, directly influencing student accountability, engagement, and institutional effectiveness. In universities such as the University of Cebu, accurately and efficiently tracking student attendance is necessary for ensuring compliance with academic policies, facilitating timely interventions for absenteeism, and enhancing the overall quality of education.

Researchers have observed that nearly every instructor at the University of Cebu allocates approximately 5% of the grading system to student attendance. This practice highlights the vital role that attendance plays in the comprehensive evaluation of student performance. Consequently, the accuracy and reliability of attendance records are essential not only for administrative purposes but also for ensuring fairness and consistency in the grading process.

Despite the availability of various attendance tracking methods, many educational institutions, including the University of Cebu, continue to face significant challenges in student attendance monitoring, specifically:

- **Inefficiency in manual attendance processes**, resulting in the loss of valuable instructional time due to roll calls and paper-based logs.
- **Inaccuracy in attendance records** is caused by human error and fraudulent practices, such as proxy attendance.
- **Lack of real-time attendance data**, limiting the ability of faculty and administrators to monitor student participation and address absenteeism promptly.
- **Security and privacy concerns** associated with current attendance systems, including unauthorized access to physical logs and insufficient protection of sensitive student information in digital platforms.

The University of Cebu exemplifies these challenges, as traditional attendance methods continue to consume instructional time, expose students to health risks from shared materials, and remain vulnerable to manipulation. Such limitations hinder the university's ability to maintain accurate records and implement timely academic support.

In response to these issues, this study seeks to address the following primary research question: How can an AI-powered facial recognition system improve the efficiency, accuracy, and security of student attendance monitoring?

FaceCheck, an AI-powered facial recognition system, offers a contactless and automated solution designed to mitigate these challenges. By utilizing advanced liveness detection technology, FaceCheck ensures that only physically present students are recorded, thereby preventing proxy attendance and enhancing data integrity. The system provides instantaneous attendance logging, enabling faculty and administrative personnel to access accurate attendance records in real-time. This capability facilitates the timely identification of absenteeism trends and supports data-driven decision-making to promote student success.

Security and privacy considerations are addressed through encrypted data storage and strict compliance with the Philippines' Data Privacy Act of 2012. Unlike conventional paper logs, FaceCheck restricts data access to authorized users and employs anonymization techniques to protect biometric information from misuse.

Through the evaluation and development of FaceCheck at the University of Cebu, this study aims to contribute to the modernization of student attendance management by demonstrating the system's potential to improve operational efficiency, enhance record accuracy, and uphold institutional accountability. The findings are expected to inform best practices for integrating innovative technologies within academic settings, thereby advancing the university's commitment to excellence and student welfare.

Objectives of the Study

General Objective

This study aims to analyze, design, and develop an AI-powered facial recognition attendance monitoring system specifically for recording student attendance and tracking faculty attendance during meetings and official events within an academic institution.

Specific Objectives:

1. Identify the current or existing attendance system in terms of:

    1.1 The process of attendance tracking; and

    1.2 The problems encountered include manual errors, proxy attendance, and inefficiencies.

2. Determine the mechanisms and technologies to be utilized for the proposed system, specifically:

    2.1. Development of a facial recognition model using OpenCV libraries for accurate participant identification;

    2.2. Integration of the facial recognition model into a backend system for automatic recording of attendance with identity verification, timestamps, and real-time processing;

    2.3. Design of a user-friendly interface that enables authorized personnel to manage attendance records, monitor recognition results, and perform error correction and manual overrides;

    2.4. Provision of real-time dashboards and analytical tools for administrators and faculty, enabling access to attendance summaries, daily logs, absentee notifications, and reports.

3. Identify the best features of the proposed AI-powered facial recognition attendance monitoring system for both classroom and faculty-related activities.

4. Evaluate the proposed system's level of acceptability and effectiveness by examining its accuracy in facial recognition, the efficiency of attendance processing, the system's overall usability, and its contribution to reducing administrative tasks and preventing instances of fraudulent attendance within academic environments.

Scope and Limitations of the Study

This study is focused on the creation and evaluation of FaceCheck, a facial recognition system powered by artificial intelligence intended to automate attendance monitoring for students and faculty within the University of Cebu.

Scope of the Study

The following activities and features define the boundaries of the study:

- Developing a facial recognition model using deep learning techniques, particularly through a Convolutional Neural Network (CNN).
- Use and testing of the system within selected classrooms and faculty offices under supervised conditions.
- Real-time attendance detection using standard webcams connected to laptops and desktop computers, with secure storage of attendance data.
- A four-month testing period, during which the system's usability, interface, and user feedback were gathered and evaluated.
- Basic security protocols should be included, such as protection against impersonation and simple safeguards for maintaining data confidentiality.
- Configuration of system rules and functions to align with the internal attendance procedures of the University of Cebu.

Limitations of the Study

Despite its focused scope, the study is subject to several limitations:

- The system is tailored for development at the University of Cebu and has not been adapted or validated in other institutions.
- It is restricted to desktop and laptop environments and does not support mobile platforms or handheld devices.
- Environmental conditions such as poor lighting, extreme facial angles, or substandard webcam quality may influence the accuracy of facial recognition.
- The system does not incorporate biometric technologies like fingerprint scanning or eye recognition.

- Broader legal and ethical concerns related to biometric data collection are not addressed in detail, aside from meeting basic privacy considerations.

- Integration with external software platforms, including Human Resource Management Systems (HRMS) and Learning Management Systems (LMS), is beyond the current development scope.

Significance of the Study

This study focuses on addressing common challenges in student attendance monitoring at the University of Cebu by introducing FaceCheck, an AI-powered facial recognition system. The system aims to enhance the recording of student attendance—making the process more efficient, accurate, and secure. Its significance is reflected across several key stakeholders:

Students. This system offers a contactless and convenient way to log attendance. It eliminates the need for manual sign-ins or physical ID verification, ensuring that only students who are physically present are marked as such. This helps promote fairness, discourage proxy attendance, and minimize classroom interruptions.

Faculty. Faculty members benefit from reduced time spent on manually taking attendance. The system allows instructors to focus more on teaching and interacting with students. Additionally, having access to accurate and real-time attendance data makes it easier to identify students who may need academic support.

Faculty Administrators. The system facilitates improved attendance tracking by maintaining a centralized and reliable database of student attendance records. This helps improve monitoring processes, simplifies documentation, and allows for timely interventions when needed. It also helps enforce attendance policies more consistently.

University of Cebu. The adoption of FaceCheck reflects the University of Cebu's ongoing commitment to innovation and academic improvement. By integrating a modern, AI-powered attendance system, the university takes a step toward enhancing operational efficiency and strengthening its student monitoring processes. This initiative also supports the university's efforts to foster a secure, data-driven, and technology-enabled academic environment that aligns with the evolving needs of higher education.

Future Researchers. This study can serve as a foundation for future research focused on enhancing AI-based attendance systems in academic settings. Future researchers may explore improvements in facial recognition accuracy, integration with mobile platforms or school portals, and analysis of long-term impacts on student attendance behavior. It also opens the door to examining ethical considerations and privacy implications of biometric data use in education.

Definition of Terms

This section provides definitions for important terminology used in the study to guarantee understanding and clarity of the concepts and technology employed in the development of the FaceCheck system.

Artificial Intelligence (AI) - Refers to the capacity of computer programs or devices to do tasks like comprehension, learning, and decision-making that often call for human thought.

Backend System - Refers to the part of a software application that works behind the scenes to manage data, perform calculations, and handle storage, without direct interaction from the user.

Convolutional Neural Network (CNN) - Refers to a type of computer model designed to analyze images by recognizing patterns, often used to identify faces or objects in pictures.

Data Privacy - Refers to the process of safeguarding private data to prevent unauthorized access or disclosure.

Data Storage - Refers to the way digital information is saved and kept so it can be accessed later when needed.

Encrypted - Refers to the method of changing data into a secret code to keep it safe from unauthorized viewing or use.

FaceCheck - Refers to the facial recognition tool created in this project that uses AI to automatically and securely record student attendance.

Facial Recognition - Refers to a technology that identifies a person by examining the distinct features of their face.

Human Resource Management Systems (HRMS) - Refers to the software used by businesses to arrange and handle employee data, including attendance, pay, and documentation.

Learning Management Systems (LMS) - Refers to online platforms that schools or organizations use to provide and track courses and learning activities.

Liveness Detection - Refers to a security feature that checks if the face presented to the system belongs to a live person, not a photo or video.

Manual Attendance - Refers to the traditional way of taking attendance by calling names or signing a sheet, which can be slow and prone to mistakes.

Philippines' Data Privacy Act of 2012 - Refers to the law in the Philippines that sets rules for how personal data should be collected and protected to respect people's privacy.

Proxy Attendance - Refers to the dishonest act of one person marking attendance for someone else who is not present.

Real-Time - Refers to the immediate updating and processing of information as events happen, without delay.

Spoofing - Refers to attempts to fool a facial recognition system by using fake images, masks, or videos.

Standard Webcam - Refers to a common camera connected to or built into a computer, used here to capture images of faces.

Usability Testing - Refers to the process of watching how actual users interact with a system to make sure it is easy and effective to use.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Review of Related Literature

In this chapter, important scholarly publications and current developments related to the creation of FaceCheck—an AI-powered face recognition system designed to automate attendance tracking—are examined. The literature is classified into international and local sources, with a focus on primary themes such as computer vision, secure attendance documentation, database management, real-time analysis, user interface design, and system usability.

Foreign Literature

Given how quickly artificial intelligence (AI) is developing, computer vision has revolutionized attendance monitoring systems worldwide, driving a shift from manual and biometric approaches to automated, AI-driven solutions. Dang (2023) proposed an innovative attendance system that utilizes advanced facial recognition methods grounded in Convolutional Neural Networks (CNNs). This system exhibited high levels of accuracy and resilience under varying environmental conditions, enabling real-time attendance monitoring and automatic logging that significantly minimizes human error and fraudulent registrations.

Building on this groundwork, Kumar et al. (2025) developed an AI-driven facial recognition attendance system explicitly tailored for workforce management. Their method combines advanced face detection algorithms with secure database management to avert proxy attendance and uphold data integrity. The system's scalability and flexibility make it suitable for a variety of organizational contexts, including educational facilities, emphasizing its extensive applicability.

In a similar vein, Nguyen-Tat, Bui, and Ngo (2024) introduced a design science methodology that merges computer vision and facial recognition to automate attendance management in human resources. For large-scale implementations, Motwani et al. (2025) proposed an automated attendee recognition system designed for social events and conferences. By utilizing AI-powered facial recognition, their system streamlines check-in processes and instant attendance registration, significantly reducing manual processing time while enhancing data accuracy and security.

Rao (2022) introduced AttenFace, a real-time facial recognition attendance system that incorporates liveness detection to reduce spoofing attempts. This system integrates smoothly with backend databases, ensuring safe, tamper-resistant attendance records. The research emphasizes the necessity of fusing advanced AI algorithms with sturdy system structures to achieve dependable and secure attendance tracking.

Furthering classroom-specific innovations, Aggarwal et al. (2024) unveiled ClassVision, an AI-driven attendance system that employs deep learning models for accurate face detection and recognition. ClassVision offers an intuitive user interface, real-time attendance updates, and comprehensive analytics, enabling effective classroom management and informed decision-making.

In addition to these advancements, Hasini et al. (2025) introduced an AI-based innovative attendance management system that stresses accuracy, user-friendliness, and security. Their system integrates facial recognition modules, encrypted database storage, and automated report generation, demonstrating how AI can optimize attendance processes while protecting sensitive information.

Together, these international studies highlight the transformative effects of AI and facial recognition technologies on attendance monitoring. They consistently emphasize the significance of real-time data capture, secure and encrypted database management, fraud deterrence through liveness detection, and the integration of robust system architectures to maintain system reliability across diverse operational environments.

Local Literature

In the Philippines, studies show a growing adoption of biometric and AI-driven attendance systems to overcome the limitations of conventional manual methods. Ani˜Non et al. (2020) examined a classroom attendance system that integrated Radio Frequency Identification (RFID) with a web-based platform. While this approach reduced manual mistakes, the research identified challenges such as card sharing and the necessity for physical contact, which raised security and hygiene issues, prompting the pursuit of contactless alternatives.

To address these issues, Domingo and Ladia (2024) developed QSUM-EASYS, a facial recognition attendance system linked to a web-based monitoring interface, which was implemented at Quirino State University. Their system enables real-time facial detection and automatic attendance recording, featuring a user-friendly web interface for administrators that results in improved accuracy and reduced administrative burdens.

Reynoso and Torres (2020) introduced Tempus, an attendance monitoring system specifically designed for schools in the Philippines that utilizes facial recognition technology. Utilizing AI algorithms for facial detection and recognition along with secure data storage, Tempus effectively addresses challenges related to proxy attendance and inaccuracies in manual data entry, thus enhancing the reliability of attendance records.

Furthering the shift toward contactless solutions, Sunico, Gonzales, and Argana (2023) developed a non-contact attendance monitoring system for chartered institutions using facial recognition technology. Their study emphasizes the hygiene advantages of contactless methods and highlights the importance of real-time monitoring and data synchronization to uphold accurate and up-to-date attendance records.

Going beyond simple attendance tracking, Bermudez et al. (2024) investigated AI-driven enhancements in university computer labs, integrating sit-in reservations and monitoring systems to enhance efficiency. Their results showcase the broader applications of AI in resource management and institutional oversight, complementing attendance systems by boosting operational effectiveness.

The local literature also notes the challenges linked with facial recognition technologies, such as inconsistent lighting conditions and hardware constraints that can affect accuracy. These challenges underscore the necessity for robust algorithm development and adaptable system frameworks to ensure reliable performance in diverse settings.

The reviewed literature, comprising both local and foreign studies, converges on the essential role of AI-enhanced facial recognition systems in revolutionizing attendance monitoring. Major themes include the application of sophisticated deep learning algorithms for accurate facial detection and recognition, the implementation of secure and encrypted database management to protect sensitive attendance data, and the development of real-time monitoring capabilities to enable immediate attendance logging and reporting.

Additionally, effective user interface design and overall system usability are crucial for fostering acceptance among students and faculty, thereby reducing administrative workloads and enhancing operational efficiency.

Despite significant progress, ongoing challenges such as environmental variability, spoofing attacks, and hardware limitations remain areas that require further enhancement. These insights inform the development of FaceCheck, which aims to offer a scalable, secure, and user-friendly attendance tracking solution tailored for the University of Cebu.

Review of Related Studies

To better understand the existing approaches to digital attendance tracking, it is important to examine systems currently in use across different sectors. Various solutions implement biometric or digital check-in methods—such as facial recognition, fingerprint scanning, or QR-based verification—and are often deployed in corporate or government settings. While many of these platforms are highly capable, they are not always designed with the unique needs of educational institutions in mind. This section reviews selected systems to extract key insights and identify features that informed the development of FaceCheck, a facial recognition-based attendance tracking solution tailored for the University of Cebu.

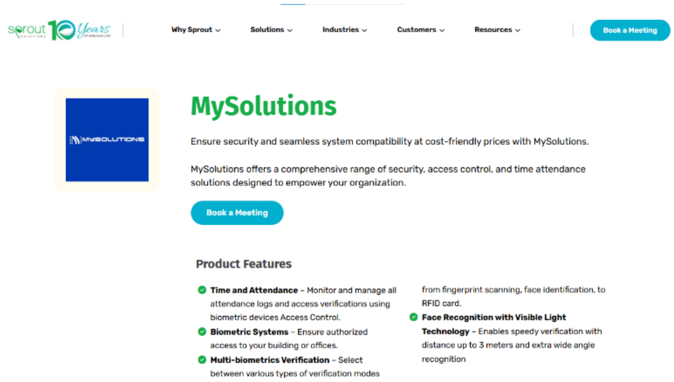

Figure 1: **Sprout MySolutions**

Sprout MySolutions is a hybrid attendance platform that incorporates fingerprint scanning, RFID access cards, and visible-light facial recognition. As part of a broader human resource and payroll solution, this system captures attendance logs in real-time. Its biometric terminal can store thousands of face templates and is designed for use in office environments. However, this system requires proprietary hardware, which increases implementation costs. More importantly, its design and feature set are geared toward enterprise environments and do not accommodate the specific scheduling needs of academic institutions. It lacks features such as class-section scheduling, roster imports, and academic-based reporting. In contrast, FaceCheck avoids proprietary hardware altogether by running on standard laptops or desktops equipped with webcams.It offers functionality specifically built around the structure of academic institutions, handling student attendance according to class schedules.

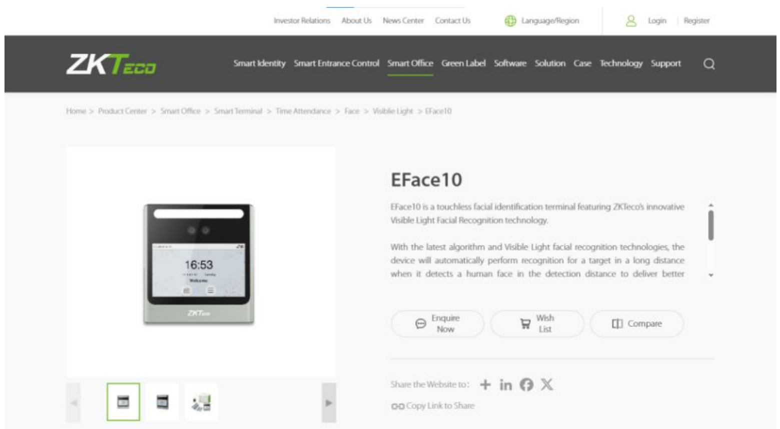

Figure 2: **ZKTeco’s EFace10**

ZKTeco’s EFace10 is a biometric attendance terminal that supports facial recognition, fingerprint scanning, and optional card access. It uses a deep-learning algorithm to verify faces, even under conditions like mask-wearing, and is capable of handling high volumes of attendance logs. With its touchscreen interface and robust verification capabilities, it is widely used in timekeeping operations. Nevertheless, it shares a limitation common to many commercial systems—it is hardware-dependent and priced at a level that may be inaccessible to schools with limited funding. Furthermore, EFace10 outputs raw logs that require further processing, and it does not provide tools for academic attendance tracking or analytics. By comparison, FaceCheck removes the need for hardware procurement and offers ready-to-use features suitable for classrooms and academic departments, streamlining the attendance process for students.

Figure 3: **eLeave Philippines**

eLeave Philippines offers a facial recognition-based attendance system that works in both standalone and cloud-based modes. It tracks time-ins and time-outs using infrared cameras and integrates with leave management systems. Although it is effective in business settings, the system relies on specialized hardware that is not cost-efficient for academic institutions. Additionally, it lacks the flexibility to handle class schedules or monitor student attendance. The focus of eLeave is squarely on employee attendance and HR management, limiting its relevance for schools. FaceCheck was designed to address this gap by eliminating the need for custom hardware and supporting attendance workflows specific to educational environments.

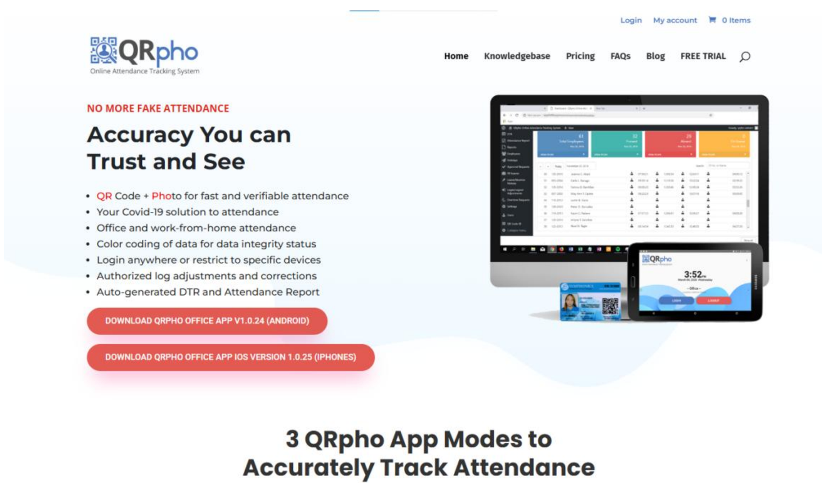

Figure 4: **QRpho**

QRpho is a QR code-based attendance system used in some government agencies and local offices. Users scan a dynamically generated QR code using their mobile phones to log their attendance. While it offers a fast and accessible check-in process, it does not include any form of biometric verification, making it vulnerable to proxy attendance. Furthermore, it does not offer automation features that are useful in academic settings. Unlike QRpho, FaceCheck utilizes facial recognition to verify that the person checking in is a registered individual.

Comparative Matrix/Competitor Analysis

The reviewed systems reveal several recurring challenges faced by institutions when adopting digital attendance technologies. Most of the commercial solutions require dedicated hardware, which raises both the initial cost and ongoing maintenance efforts. Additionally, many of these platforms were originally developed for corporate or government use, and as such, they do not accommodate the dynamic class schedules, academic calendars, and student grouping structures present in universities. Moreover, the reviewed systems generally lack the flexibility to manage students in terms of class-based attendance and academic reporting.

Through this analysis, it becomes clear that while existing systems may be effective in business settings, they fail to meet the particular requirements of academic institutions sufficiently. FaceCheck aims to fill this gap by offering a flexible, low-cost, and education-focused alternative. It was developed to support student attendance tracking, work seamlessly on commonly available devices, and align with the day-to-day operations of the University of Cebu.

Table 1

COMPARATIVE MATRIX

Features/ Functions	**Sprouts MySolutions**	**ZKTeco EFace10**	**eLeave Philippines**	**QRpho**	**FaceCheck**	**Total:**
Uses Facial Recognition	✓	✓	✓	✗	✓	80%
Supports Standard Webcams	✗	✗	✗	✗	✓	20%
Works on Standard Laptop or Desktop	✗	✗	✗	✓	✓	40%
Real-time Attendance Logging	✓	✓	✓	✓	✓	100%
Handles Academic Scheduling	✗	✗	✗	✗	✓	20%
Provides Academic Attendance Reports	✗	✗	✗	✗	✓	20%

Prevents Proxy Attendance	✓	✓	✓	✗	✓	80%
User-Friendly Interface For Education	✗	✗	✗	✗	✓	20%
Cost-Effective (No Proprietary Hardware)	✗	✗	✗	✓	✓	20%
Total:	40%	40%	30%	30%	100%	

CHAPTER III

DESIGN AND METHODOLOGY

Research Design

Method

This study employs a developmental, applied research design aimed at building and evaluating a prototype attendance monitoring system, FaceCheck, powered by AI-driven face recognition. A mixed-methods approach is utilized: quantitatively, system performance (recognition accuracy and processing time) and user satisfaction are assessed through structured surveys; qualitatively, open-ended feedback and interviews capture participants' experiences, perceptions of privacy, and ethical considerations.

Flow of the Study

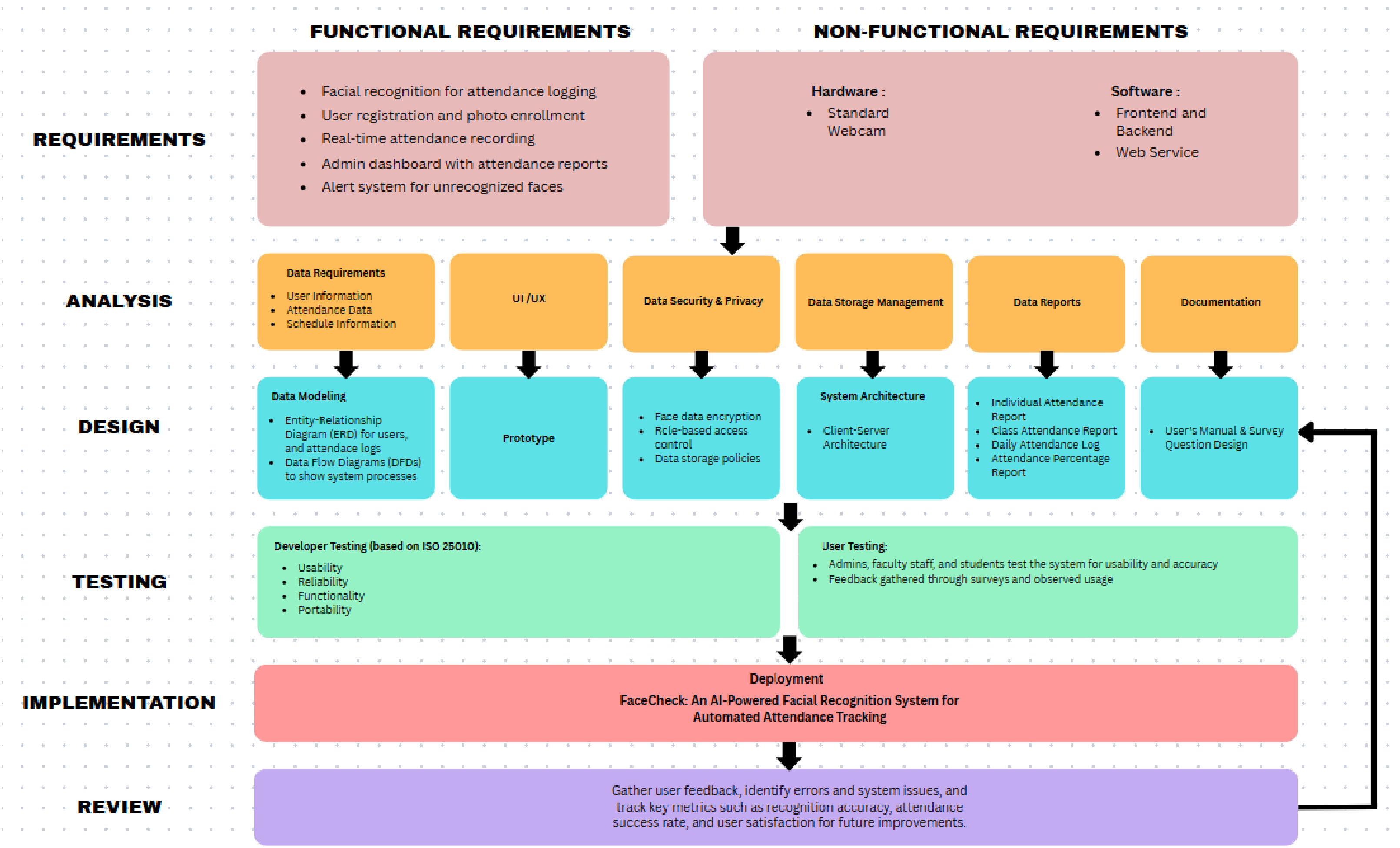

Figure 5: **Flow of the Study**

Research Environment

The study was conducted within the academic premises of the College of Computer Studies at the University of Cebu (Main Campus), where the environment closely reflects a real-world use case for the proposed facial recognition attendance system. The academic environment provided a structured yet flexible setting that allowed the researchers to test the system across various scenarios, including morning and afternoon classes, different lighting conditions, and among a diverse group of students. The presence of faculty members who actively monitored and recorded attendance also allowed for the seamless integration of the system into existing administrative routines. This real-time testing and simulation enabled the research team to observe natural user behavior, system efficiency, and potential technical issues in a context highly relevant to the project's goals. The environment also supported open feedback from users, which contributed to the continuous refinement and validation of the FaceCheck system.

Respondents

The respondents in this study were comprised of two key groups: students and faculty members from the College of Computer Studies. The method of purposive sampling was used to choose 30 participants who were directly involved in academic activities where attendance monitoring is a routine requirement. Of these, 25 were undergraduate students regularly attending scheduled lectures, while 5 were instructors responsible for managing class rosters and verifying attendance records.

The rationale for selecting these respondents lies in their direct interaction with traditional attendance systems, giving them the insight necessary to evaluate the usability, efficiency, and practical advantages of FaceCheck. The diversity in roles allowed the researchers to capture perspectives from both end-users (students) and system administrators (instructors), ensuring a comprehensive assessment of the system's performance. Students provided feedback on the facial recognition process, user interface, and system responsiveness, while faculty members evaluated the back-end functionalities, reporting features, and accuracy of attendance logs. Participants were briefed on the nature of the study, and consent was secured before any interaction with the system.

Research Instrument

To assess user experience, perceptions, and expectations regarding the FaceCheck system, the researchers developed a structured, researcher-made survey questionnaire. The instrument gathered both quantitative and qualitative data from students and faculty members through a combination of multiple-choice and open-ended questions.

The questionnaire collected background information, including the respondent's role and commonly used devices. It also explored their current experiences with existing attendance systems, including levels of satisfaction, time efficiency, and common issues like proxy attendance and technical difficulties. Furthermore, the survey gauged participants' attitudes toward facial recognition technology, covering aspects such as privacy, comfort, reliability, and preferences for system features.

Through capturing a broad range of user feedback, the instrument was a valuable tool for evaluating the system's relevance, usability, and potential acceptance within an academic environment.

Research Procedure

Data Gathering

To gather relevant data on current attendance practices and the perceptions of potential users toward a facial recognition-based attendance system, the researchers conducted a structured survey. This survey served as the primary data collection instrument for the study. It was intended to collect both qualitative and quantitative information from students and faculty members within the institution.

The survey was administered through Google Forms for ease of distribution and accessibility. The link was shared via institutional communication platforms, including official group chats, emails, and academic social media groups. The questions were a combination of multiple-choice, scaled, and open-ended items, enabling the researchers to gather both statistical insights and subjective feedback. No personally identifiable information (PII) was collected unless voluntarily shared in open-ended responses. The collected data was then organized and prepared for analysis to inform the development and validation of the proposed FaceCheck system.

Treatment of Data

The data collected from the survey was organized and analyzed to extract relevant insights regarding the institution's existing attendance practices and the potential for adopting a facial recognition system. Quantitative data, such as multiple-choice and scaled responses, were tabulated and summarized using descriptive statistics, including frequencies and percentages. These were used to identify trends in satisfaction levels, time spent on attendance, and the prevalence of issues such as proxy attendance or technical difficulties.

Qualitative data gathered from open-ended responses were reviewed and thematically coded to highlight recurring suggestions, concerns, and expectations from potential users of the system. All responses were treated with strict confidentiality. The aggregated results served as a basis for identifying user needs, validating system feasibility, and informing the design and refinement of the FaceCheck attendance system.

Ethical Considerations

The development and testing of the FaceCheck system adhere to ethical standards for handling biometric data in academic research. All participants involved in the testing phase are fully informed of the nature and purpose of the study, including the use of facial recognition technology. Informed consent is acquired, and participation is voluntary before system registration. The system is designed to minimize privacy risks by avoiding the storage of raw facial images and instead relying on encrypted facial encoding for recognition. Access to biometric and attendance data is restricted to authorized personnel only and protected using secure authentication and encryption methods. No data will be shared with third parties, and the system will not be used for any purpose beyond the scope of this capstone project. Ethical considerations such as consent, data security, transparency, and the right to opt-out are strictly upheld throughout the research process.

Software Engineering Methodology

For the development of FaceCheck: An AI-Powered Facial Recognition System for Automated Attendance Tracking, the Iterative Software Development Methodology is adopted. This methodology is particularly effective for projects that require progressive refinement, ongoing testing, and incorporation of stakeholder feedback. Given the complexity and evolving nature of artificial intelligence and facial recognition technologies, the iterative model enables the system to be developed incrementally through repeated development cycles.

Each iteration in the development process includes distinct phases such as planning, requirements gathering, analysis and design, implementation, testing, and evaluation. After each cycle, feedback is used to reassess goals, enhance system features, and resolve any identified issues.

This cyclic approach enables the gradual improvement of key modules, such as face detection, user authentication, and attendance recording, ensuring that the system matures with each iteration.

In employing this methodology, the development team can mitigate risks early, accommodate changes in requirements, and steadily build a robust and high-performing attendance tracking system. The iterative process not only improves adaptability and quality but also promotes a more structured and manageable pathway to achieving the final system objectives.

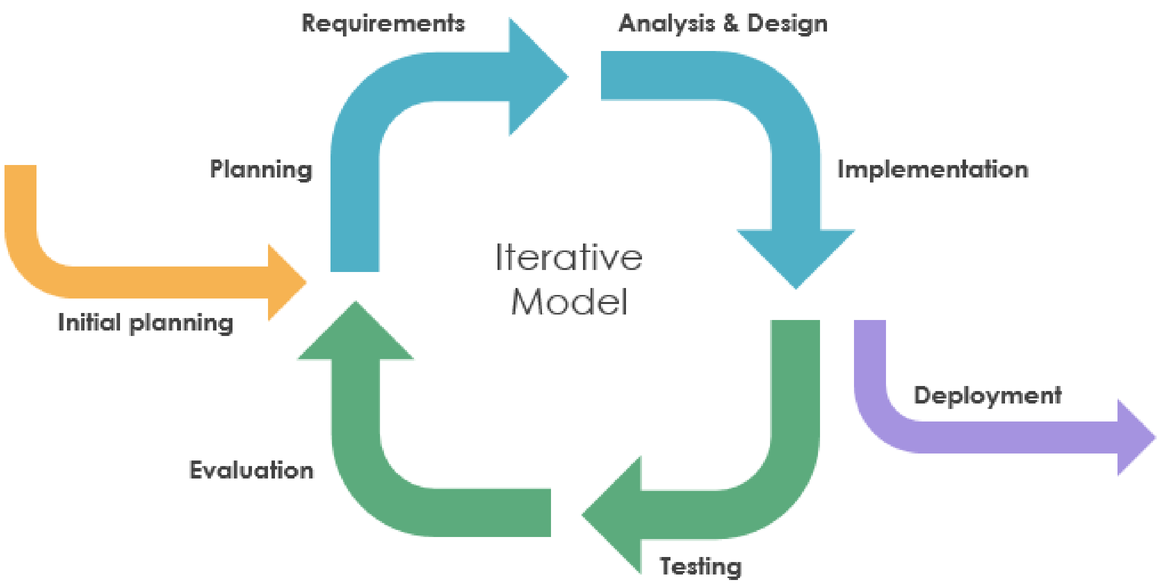

Figure 6: **Iterative Software Engineering Methodology**

Planning/Conception-Initiation Phase

This section encompasses essential components, including the business model canvas, program workflow, system architecture design, project timeline, and facial recognition model specifications. These elements serve as crucial planning and initiation tools for the proposed FaceCheck AI-powered attendance monitoring system, facilitating the systematic approach toward addressing the identified problems in traditional attendance management within educational institutions.

Business Model Canvas

This Business Model Canvas presents the strategic outline for the FaceCheck attendance system that utilizes AI-driven facial recognition specifically designed for educational establishments. The advancement of this system is backed by essential collaborators, such as manufacturers of cameras, suppliers of hardware, and providers of software libraries.

Key resources involve the AI development team, OpenCV libraries, hardware infrastructure, and database management tools. The value proposition centers on secure and contactless attendance, preventing proxy attendance, and providing real-time analytics and reporting.

FaceCheck fosters strong customer relationships through regular system updates, comprehensive training, dedicated support, and effective feedback mechanisms. It reaches its market through direct sales, digital platforms, and training seminars. The primary customer segment is educational institutions seeking modernized attendance solutions.

The cost structure includes research and development, programming, and hardware infrastructure, while revenue streams are generated from subscription fees, support services, and update contracts.

Table 2

BUSINESS MODEL CANVAS

<table>
<tr>
<td rowspan="2">KEY PARTNERS
● Camera Manufacturer
● Local Hardware Supplier
● Software Libraries Holder</td>
<td>KEY ACTIVITIES
● AI Model Development
● Facial Recognition Optimization
● Develop User Interface
● System Integration & Testing
● Market System</td>
<td rowspan="2">VALUE PROPOSITION
● To develop a secure & contactless attendance
● To integrate a proxy attendance prevention
● To provide real-time analytics & reporting</td>
<td>CUSTOMER RELATIONSHIP
● Customer Support
● System Updates
● User Training
● Hardware Facilities Recommendation
● Feedback Collection</td>
<td rowspan="2">CUSTOMER SEGMENT
● Educational Institutions
● Corporate Businesses</td>
</tr>
<tr>
<td>KEY RESOURCES
● AI Development Team
● OpenCV Libraries
● Hardware Infrastructure
● Developments Tools & Software
● Database Management</td>
<td>CHANNELS
● Direct sales to schools
● Campus promotions
● Website & social media
● Training Workshops & Seminars</td>
</tr>
<tr>
<td colspan="3">COST STRUCTURE
● Research and Development Expenses
● Development and Programming Costs
● Hardware Infrastructure Costs</td>
<td colspan="2">REVENUE STREAM
● Subscription Fees
● Training & Support Services
● Maintenance & Update Contracts</td>
</tr>
</table>

Program Workflow

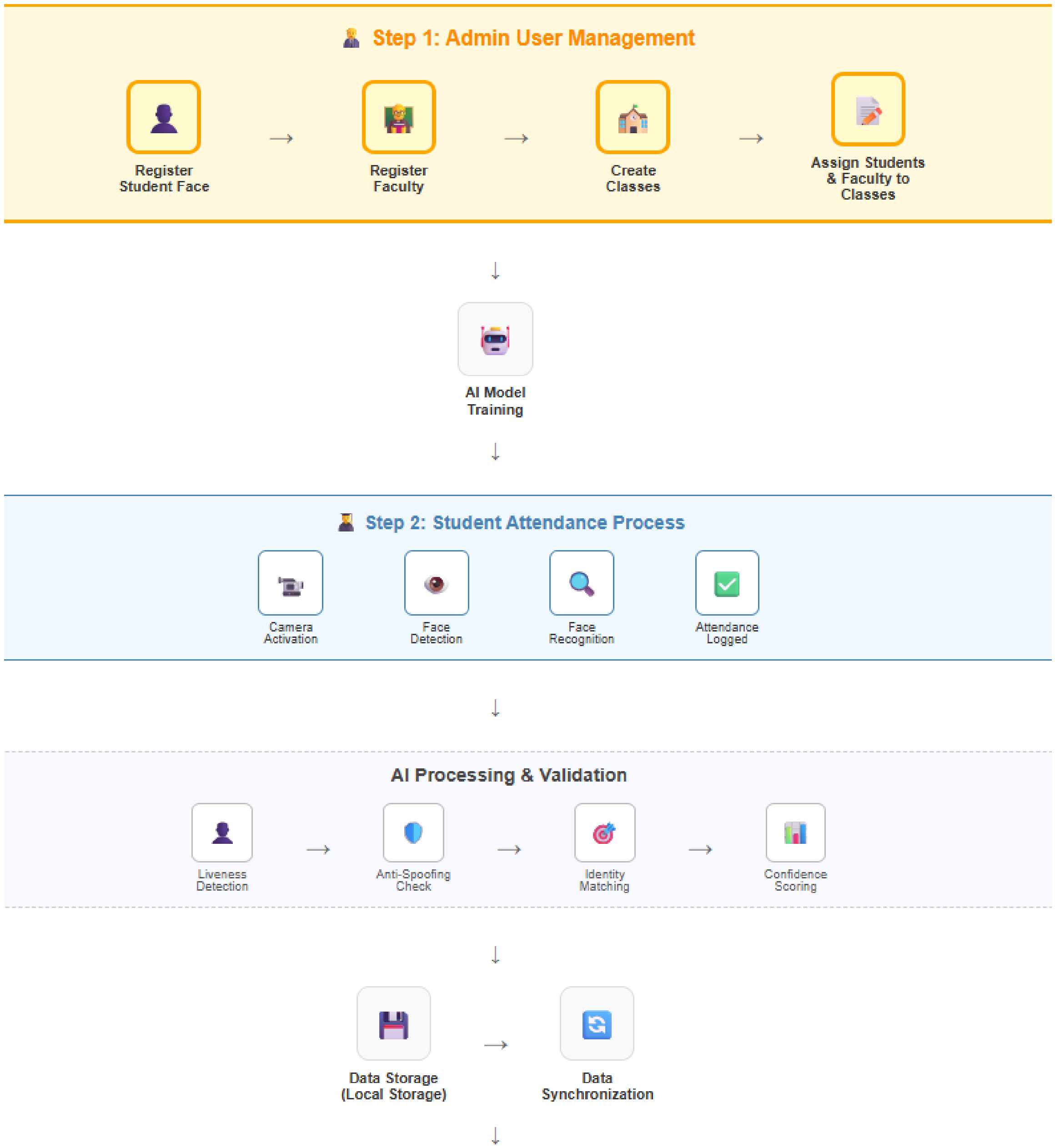

Figure 7.1: **Program Workflow(Admin/Faculty)**

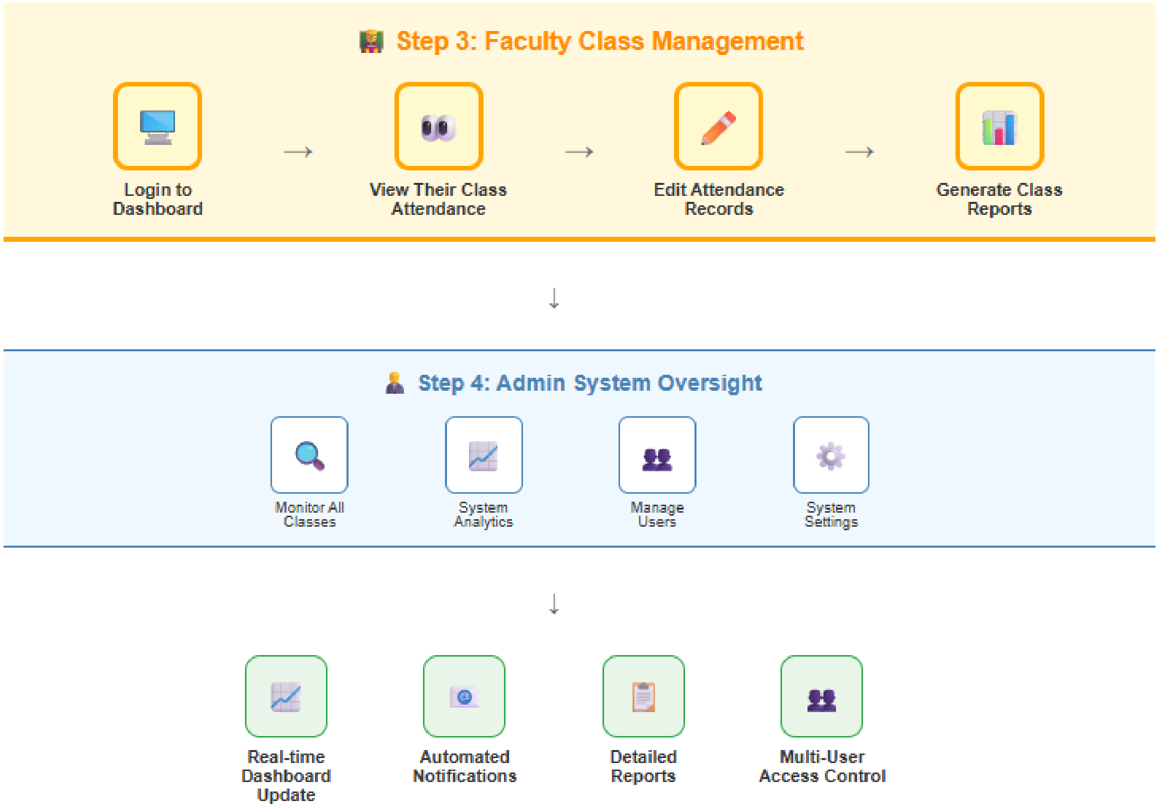

Figure 7.2: **Program Workflow (Admin/Faculty)**

This workflow diagram illustrates how to manage attendance monitoring and user administration within the FaceCheck system. The process starts with administrators registering student faces and faculty members, then creating classes and assigning students and faculty to classes. Once set up, the system undergoes AI model training, enabling students to access the attendance process. Students can perform attendance logging through face activation, detection, face recognition, and attendance logging, which undergoes AI processing and validation.

The diagram breaks down these tasks further: AI processing and validation involves liveness detection, anti-spoofing measures, identity matching, and confidence scoring before data storage and synchronization; faculty class management allows instructors to log into dashboards, view their class attendance, edit attendance records, and generate class reports; and administrative system oversight enables monitoring of all classes, system analytics, managing users, and system settings. The system also features additional capabilities to handle real-time dashboard updates, automated notifications, detailed reports, and multi-user access control functionalities.

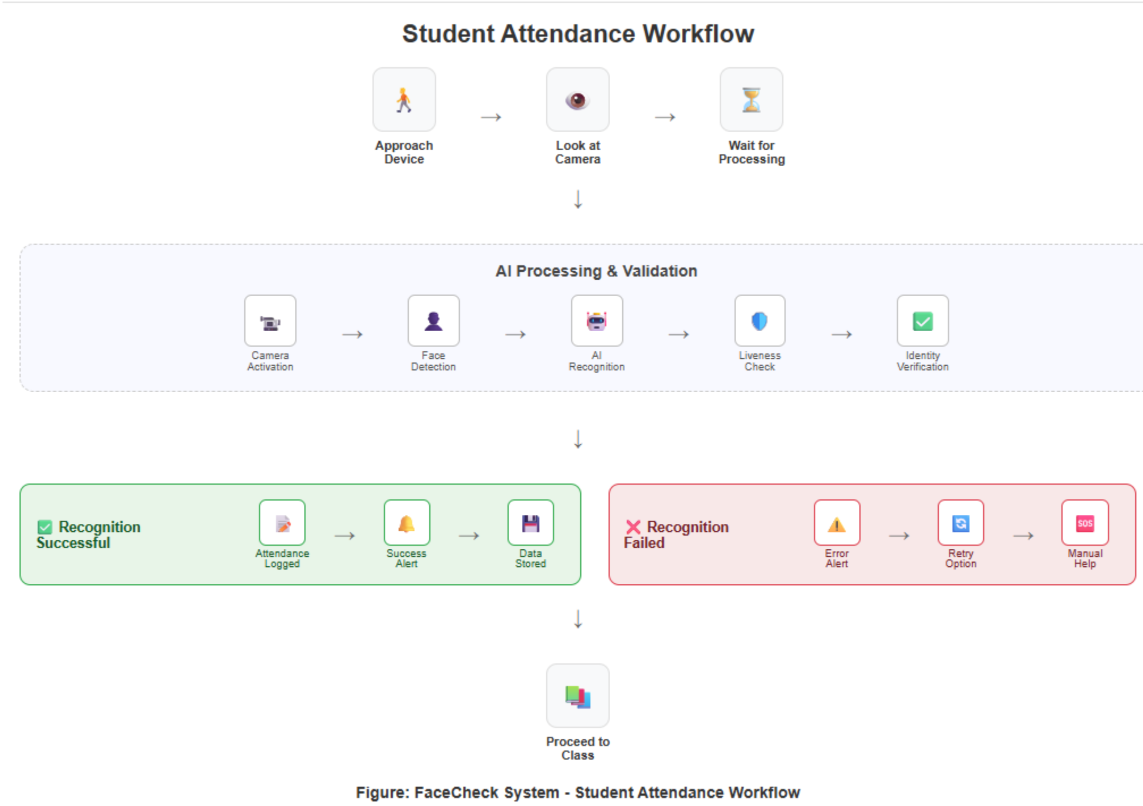

Figure 8: **Program Workflow (Student)**

This workflow diagram illustrates how to process student attendance within the FaceCheck system. The process starts with students approaching the device and looking at the camera, then waiting for processing. Once initiated, the system undergoes AI processing and validation through camera activation, face detection, AI recognition, liveness check, and identity verification to ensure accurate attendance recording.

The diagram breaks down the outcomes into two paths: recognition successful leads to attendance logged, success alert, and data stored, while recognition failed triggers an error alert, retry option, and manual help if needed. The system also features the capability to proceed with all successful attendance records to the class database for comprehensive tracking and record management.

The AI processing involved in this workflow uses facial recognition techniques based on a Convolutional Neural Network (CNN) model. When a student looks at the webcam, the system first detects the face using OpenCV's Haar Cascade or DNN-based detection methods. The detected facial region is then converted into a numerical encoding—a unique vector representation of the student's facial features. This encoding is compared against pre-stored encodings in the system database to find a match using a similarity threshold (e.g., Euclidean or cosine distance).

To enhance security and prevent spoofing, a liveness detection mechanism is integrated. This may include prompts for blinking, slight movements, or texture analysis to ensure that the input is from a live person, not a photo or video. Once both identity and liveness are confirmed, the system logs the attendance with the current timestamp and updates the class record in real-time. The entire process is automatic, contactless, and typically completed within a few seconds.

Validation Board

Table 3

STAGE 1: PROBLEM VALIDATION

	Facilitators	Attendees
Problem	• Time-consuming roll calls disrupt instructional time • Manual logs susceptible to proxy attendance • Current systems create classroom interruptions • Attendance tracking reduces effective teaching time • Proxy attendance is widespread and problematic	• System errors in current attendance methods • Inconvenience with current ID cards or biometric scanners • Health concerns with shared materials/devices • No assurance of attendance data privacy • Limited reliability of current tracking systems
Riskiest Assumptions	The problem exists and significantly impacts teaching efficiency	The problem exists and affects student experience
Success Criteria	80% PERSEVERE	70% PERSEVERE

Results and Discussions	85.2% identify time-consuming process as major challenge. 85.2% report proxy attendance as significant issue. 74% spend 4-6 minutes on attendance daily. 96.3% have observed proxy attendance (frequently + occasionally), confirming widespread nature of the problem.	66.7% are neutral to dissatisfied with current processes. 96.3% have observed proxy attendance, showing system reliability issues. 18.5% report health risks as concern. Current system reliability rated as only 55.6% "somewhat reliable" and 18.5% "unreliable."
Learning	Strong need for automation to enhance instructional time. Faculty ready to adopt AI-based solutions to address widespread proxy attendance issues.	Students value contactless, reliable technology for attendance. Current systems have significant reliability and health concern issues that need addressing.

Table 4

STAGE 2: SOLUTION VALIDATION

	Facilitators	Attendees
Solution	AI-powered facial recognition attendance system (FaceCheck) that automates roll calls, prevents proxy attendance, improves instructional efficiency, and provides real-time tracking with minimal classroom interruption.	Contactless, hygienic attendance logging system with quick and reliable facial recognition, privacy-protected biometric data handling, and user-friendly interface requiring minimal interaction.

Riskiest Assumptions	The proposed solution addresses faculty problems and will be adopted	The proposed solution addresses student concerns and will be accepted
Success Criteria	80% PERSEVERE	70% PERSEVERE
Results and Discussions	81.5% want real-time attendance dashboard feature. 66.7% prefer notifications for absences. 81.5% want manual override capability for system errors. 92.6% want user-friendly interface. Solution addresses core time-consumption and proxy attendance issues identified in Stage 1.	100% support FaceCheck adoption at University of Cebu. 33.3% are comfortable/very comfortable with facial recognition, while 51.9% are neutral (not opposed). 85.2% want data encryption for comfort. 77.8% prefer limited access to authorized users. 92.6% prioritize user-friendly interface.
Learning	The proponents can learn that faculty strongly support automation solutions when they address core pain points. Manual override capability is essential for faculty confidence. User-friendly interface and real-time features are critical for adoption success.	The positive feedback emphasizes that despite privacy concerns, students strongly support contactless solutions. Security features (encryption, limited access) are critical for acceptance. User experience is the highest priority, and universal support indicates readiness for implementation when privacy concerns are addressed.

Business Roadmap

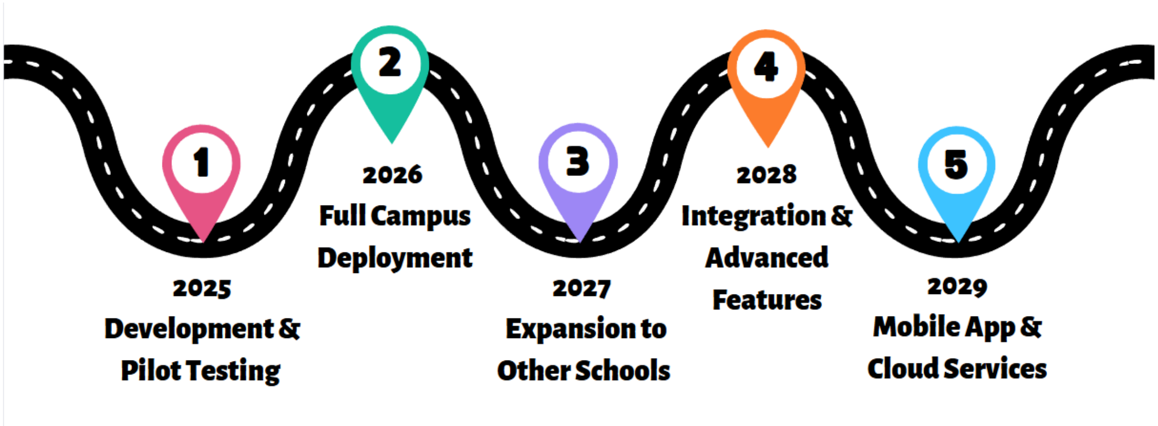

Figure 9: **Business Roadmap**

The figure illustrates the strategic business roadmap for the FaceCheck system, detailing its phased development, deployment, and growth from 2025 to 2029. This timeline outlines a systematic and sustainable approach to implementing an AI-driven facial recognition attendance system for students, which will be overseen and monitored by faculty and administrators.

In 2025, the initial development and pilot implementation of the system will occur at selected departments within the University of Cebu. This phase will prioritize establishing FaceCheck's essential functionality, which automates student attendance via facial recognition. Input from faculty and administrators will be vital for assessing system performance and identifying areas for enhancement.

By 2026, FaceCheck will be fully deployed throughout the university. Improvements such as offline attendance recording, automated report creation, and secure role-based access for faculty and administrators will be added to facilitate large-scale institutional operations.

In 2027, the system will be launched in nearby colleges and universities within Region VII. Customizable features will be developed to cater to the diverse needs of various institutional frameworks, while initiatives such as partnerships and promotional campaigns will help promote the broader adoption of these features.

In 2028, FaceCheck will begin integrating with existing student information systems to enhance data management and administration. This phase will also introduce advanced analytics capabilities, providing faculty and administrators with attendance insights to support data-driven decision-making.

Finally, in 2029, the system will broaden its accessibility through the launch of a mobile application and transition to a cloud-based infrastructure. The purpose of these improvements is to improve user experience and reduce hardware and maintenance demands, thereby making the system more accessible for institutions with limited resources.

Functional Decomposition Diagram

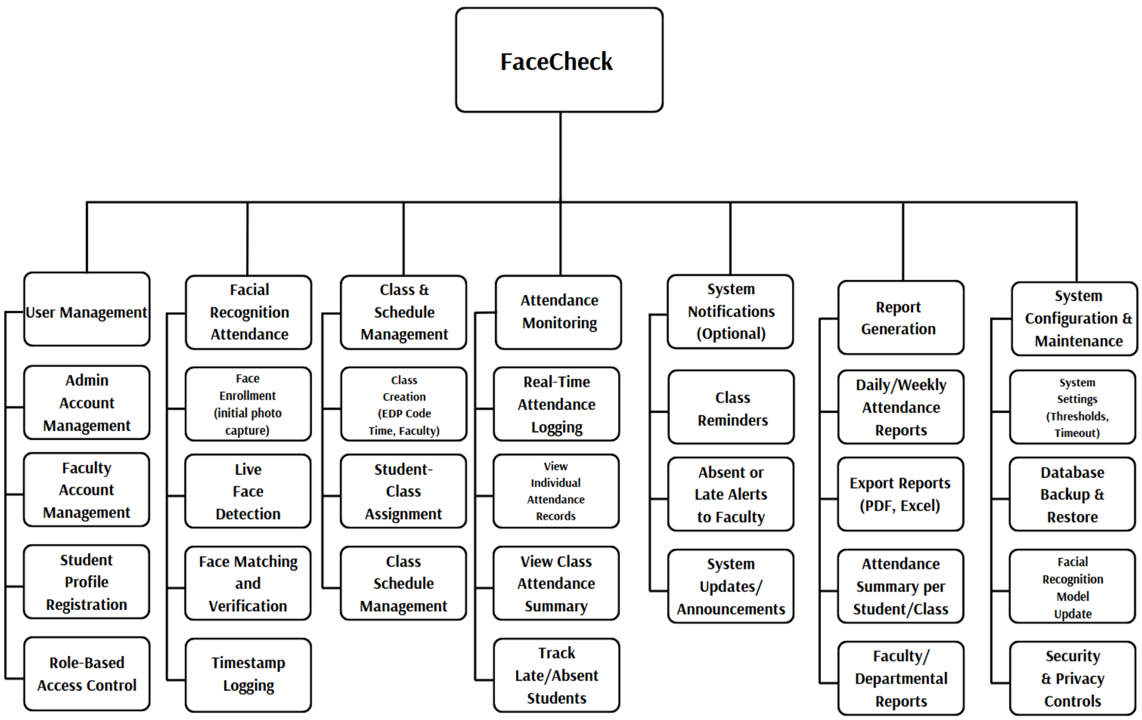

Figure 10: **Functional Decomposition Diagram**

The functional decomposition diagram for the FaceCheck Attendance System illustrates a comprehensive hierarchical breakdown of the system's core functionalities organized into seven primary modules. The system encompasses User Management (handling admin, faculty, and student account management with role-based access control), Facial Recognition Attendance (covering face enrollment, live detection, matching verification, and timestamp logging), Class and Schedule Management (facilitating class creation with EDP codes, student-class assignments, and schedule coordination), and Attendance Monitoring (enabling real-time logging, individual and class attendance tracking, and late/absent student monitoring). Additionally, the system includes System Notifications (providing optional class reminders, alerts to faculty, and system announcements), Report Generation (supporting daily/weekly reports, PDF/Excel exports, and comprehensive attendance summaries), and System Configuration and Maintenance (encompassing system settings, database backup/restore, facial recognition model updates, and security controls). This structured decomposition demonstrates the system's modular architecture, where each primary function is systematically broken down into specific sub-functions that collectively deliver an integrated biometric attendance solution.

Gannt Chart

Table 5

GANTT CHART

Iterative Gantt Chart: FaceCheck

	Planned
	Actual

Features	PLANNED ITERATION START	PLANNED ITERATIONS	ACTUAL ITERATION START	ACTUAL INTERATION	TASK COMPLETED	MAY				JUNE				JULY				AUGUST				SEPTEMBER				OCTOBER				NOVEMBER			
						1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
CHAPTER I																																	
Rationale	3	4	3	4	100%																												
Objectives	3	4	3	4	100%																												
Scope and Limitations	3	4	3	4	100%																												
Significance	3	4	3	4	100%																												
Definition of Terms	3	4	3	4	100%																												
CHAPTER II																																	
Related Literature	3	4	3	4	100%																												
Related Studies	3	4	3	4	100%																												
Comparative Matrix	3	4	3	4	100%																												
CHAPTER III																																	
Research Design																																	

Method	2	3	2	3	100%																												
Flow of the Study	2	3	2	3	100%																												
Research Environment	2	3	2	3	100%																												
Respondents	2	3	2	3	100%																												
Research Instrument	2	3	2	3	100%																												
Research Procedure																																	
Treatment of Data	2	3	2	3	100%																												
Ethical Considerations	2	3	2	3	100%																												
Software Engineering Methodology	2	3	2	3	100%																												
Planning/Conception-Initiation Phase																																	
Business Model Canva	2	3	2	3	100%																												
Program Workflow	2	3	2	3	100%																												
Validation Board	2	3	2	3	100%																												
Business Roadmap	2	3	2	3	100%																												
Functional Decomposition Diagram	2	3	2	3	100%																												
Gantt Chart	2	3	2	3	100%																												
Analysis-Design Phase																																	
Use Case Diagram	2	3	2	3	100%																												
Story Board	2	3	2	3	100%																												
Database Design	2	3	2	3	100%																												
Entity-Relationship	2	3	2	3	100%																												
Data Dictionary	2	3	2	3	100%																												

Network Design					
Network Model	2	3	2	3	100%
Network Topology	2	3	2	3	100%
Development/Construction Build Phase					
Technology Stack	4	1	4	1	100%
Software Specification	4	1	4	1	100%
Hardware Specification	4	1	4	1	100%
Program Specification	4	1	4	1	100%
List of Modules	4	1	4	1	100%
Testing Plan					
Unit Testing	4	1	4	2	100%
Integration Testing	4	1	4	2	100%
Alpha Testing	4	1	4	2	100%
Acceptance Testing	4	1	4	2	100%
Curriculum Vitae	4	1	4	2	100%
Modules					
User Management Module					
Create User Account	1	2			
Edit User Account	1	2			
Assign User Roles	1	2			
Reset Password	1	2			

Deactivate Account	1	2			
Facial Registration & Recognition Module					
Face Enrollment	2	4			
Capture Face Template	2	4			
Real-Time Face Detection	2	4			
Liveness Detection (Anti-Spoofing)	2	4			
Log Recognition Event	2	4			
Attendance Tracking Module					
Auto Timestamp Attendance	4	2			
Log Attendance to Database	4	2			
View Attendance Record	4	2			
Verify Identity on Recognition	4	2			
Prevent Duplicate Logging	4	2			
Attendance Monitoring Module					
View Daily Logs	2	4			
Generate Attendance Report	2	4			
Filter by Class/Date	2	4			
Edit/Override Attendance	2	4			
Export Attendance File (PDF/Excel)	2	4			
Class & Event Management Module					
Create Class & Event Schedule	4	2			

Assign Faculty to Classes & Events	4	2			
Enroll Students to Classes	4	2			
View Class & Event Lists	4	2			
Update Class & Event Info	4	2			
Report & Analytics Module					
Create Class Schedule	2	4			
Absence Patterns Analytics	2	4			
Monthly Attendance Graphs	2	4			
Export Report CSV	2	4			
Compare Attendance Across Classes	2	4			
System Configuration Module					
Set Attendance Rules	4	2			
Configure Email Notifications	4	2			
Adjust Recognition Sensitivity	4	2			
Change System Settings	4	2			
Manage Database Settings	4	2			
Testing					
Unit Testing	4	2			
Integration Testing	4	2			
Alpha Testing	4	2			
Acceptance Testing	4	2			

Analysis - Design Phase

This section contains use case diagrams, a storyboard, a database design, and a network design for analyzing and designing the proposed system.

Use Case Diagrams

The Use Case Diagram illustrates the interaction between the three primary actors of the FaceCheck System—Attendee, Facilitator, and Administrators—and their respective roles in the automated attendance monitoring process, which is powered by facial recognition.

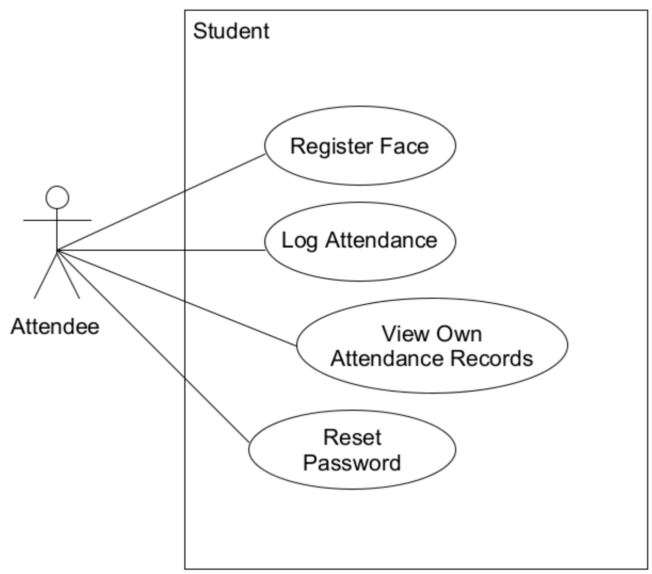

Figure 11.1: **<u>Attendee Use Case</u>**

Attendees are the end-users of the system, composed of students and faculty members who participate in classes or institutional events. Their primary functions include registering their facial data, logging their attendance in assigned sessions or events, viewing their own attendance records, and updating their account password.

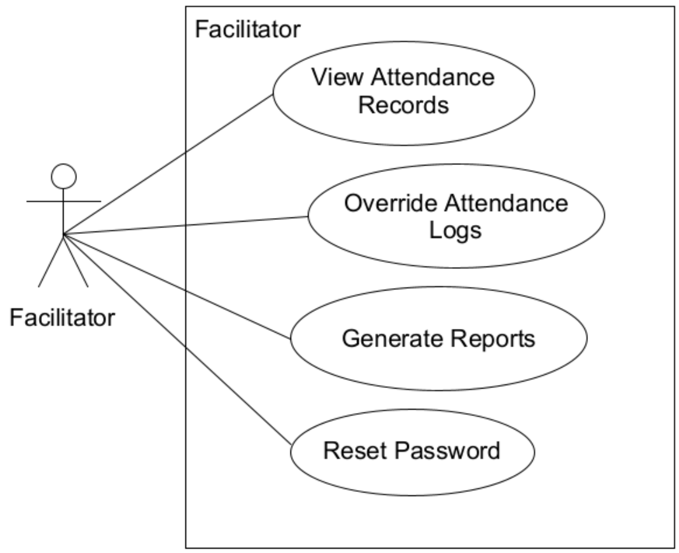

Figure 11.2: **Facilitator Use Case**

Facilitators are assigned by the administrator to manage class or event attendance. They are authorized to view attendance records, generate attendance-related reports, and override or correct attendance logs when necessary. Facilitators also have the ability to reset passwords for users under their supervision, but they do not have permission to create or assign events.

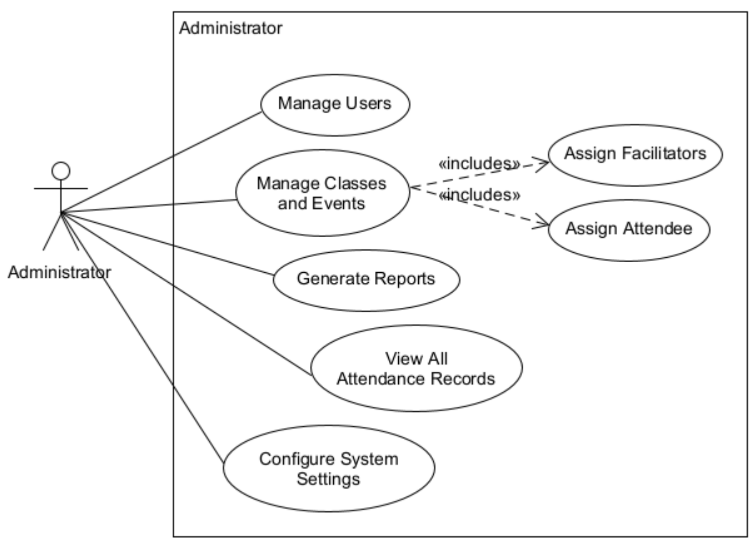

Figure 11.3: **Administrator Use Case**

Administrators oversee the overall system operation. Their responsibilities include managing user accounts, assigning faculty and students to specific classes, and configuring system settings as needed. Administrators can also generate department attendance reports to support decision-making and compliance with academic policies.

These use case structures support the FaceCheck System's goal of delivering a reliable, role-specific, and secure attendance tracking process in an educational environment.

Storyboard

A storyboard in our system is a visual representation that illustrates how users will interact with our interface. It is an essential instrument for comprehension and refining the user experience by adding a realistic context to our design process. It enables us to iterate on system flow and ensure a user-friendly interface that meets user expectations and needs.

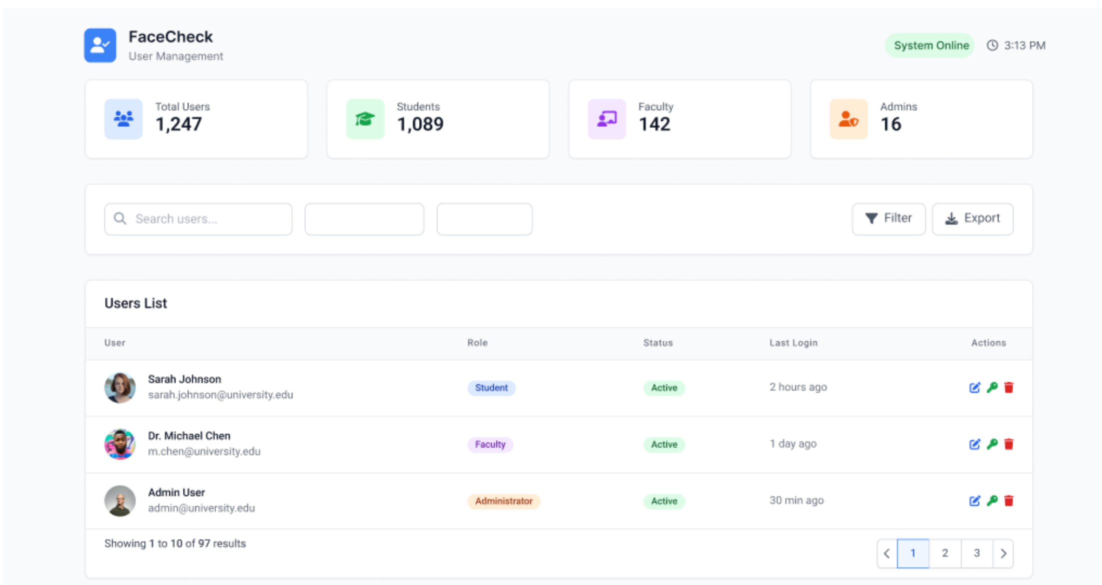

Figure 12.1: **User Management**

The interface presents an administrative dashboard that allows system administrators to manage user accounts. A user list table with search and filter functionality allows efficient browsing. Buttons like "Add User," "Edit," "Deactivate," and "Reset Password" are positioned. A modal or side panel appears when assigning user roles (Student, Faculty, Administrator), with visual cues for the access level. Emphasis is placed on security and ease of use.

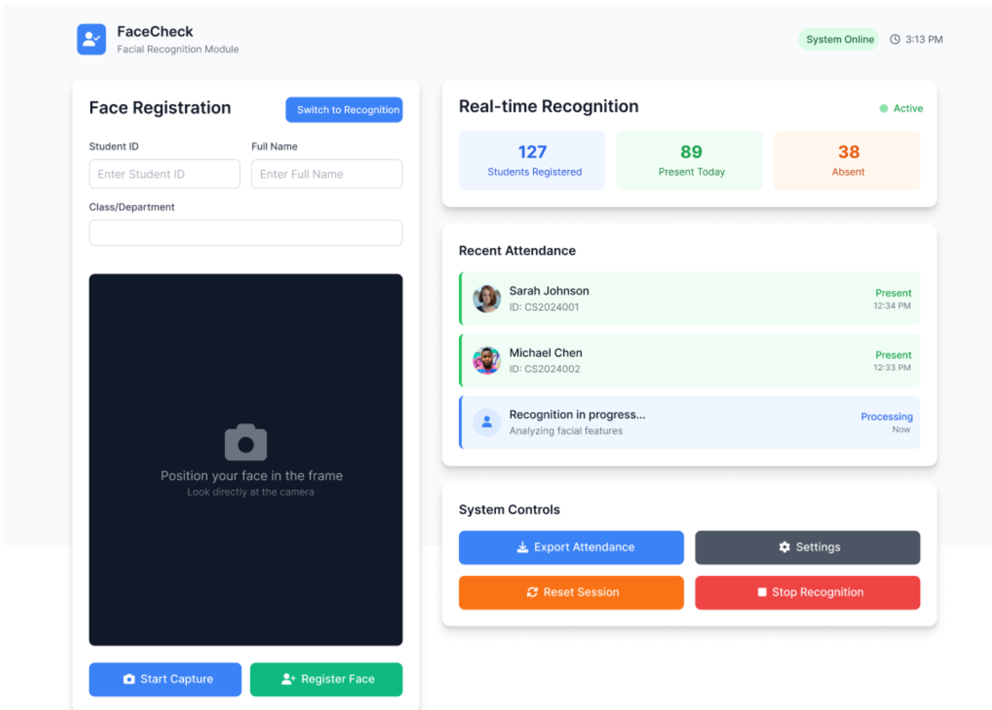

Figure 12.2: **Facial Registration**

Students are guided through a step-by-step facial enrollment process. The interface features a live webcam preview with face alignment indicators, real-time capture prompts, and success/error notifications. During recognition, the system verifies identity and logs attendance automatically. A progress/status section confirms successful check-in.

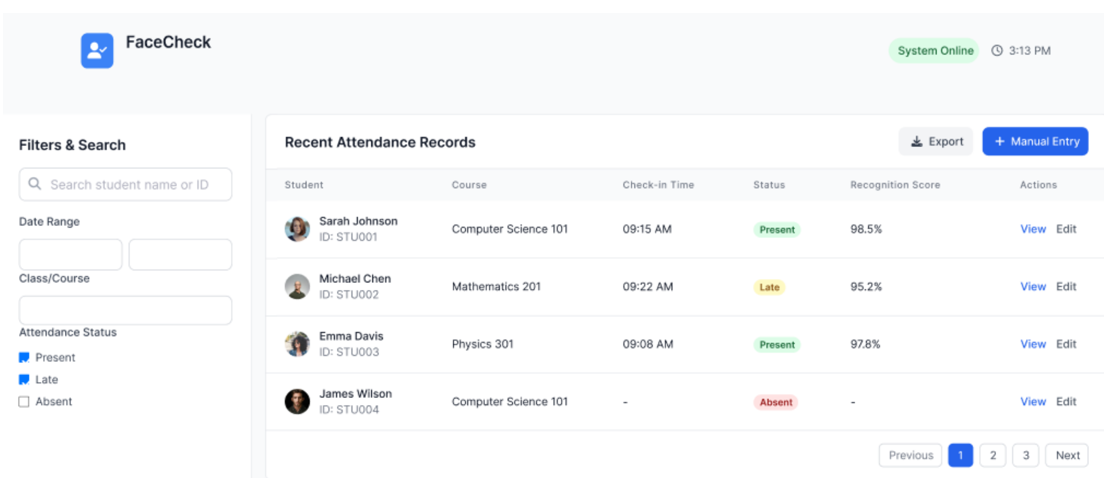

Figure 12.3: **Attendance Tracking**

Displays a timeline or tabular view of daily attendance logs captured via facial recognition. Each entry includes a student name, ID, timestamp, and attendance status. Filtering options allow users to search by date, session, or student. Data is stored automatically and updated in real time.

Faculty or administrators can view class-wise attendance summaries with color-coded indicators for present, late, or absent. Inline edit options allow authorized overrides, such as marking excused absences. A "Generate Report" button allows you to download or print session-specific data.

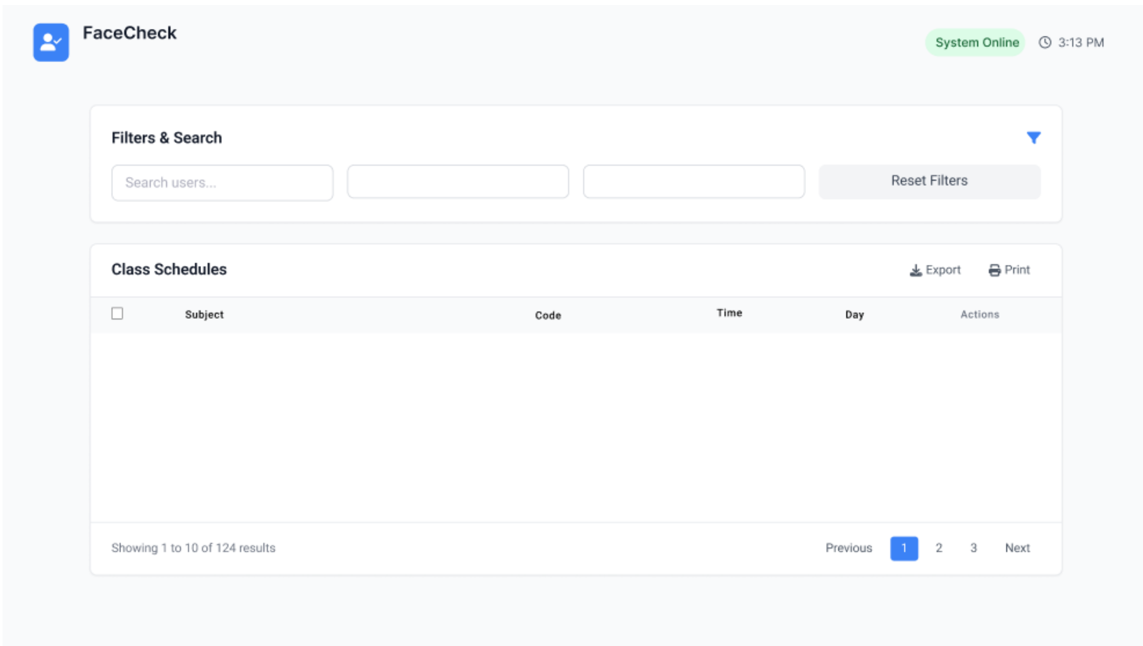

Figure 12.4: **Attendance Monitoring**

An interface for managing class lists, schedules, and assignments. A calendar or schedule view helps create and modify class times. Dropdowns or drag-and-drop features assign faculty and students to classes. Visual organization aids in managing overlapping schedules and class capacity.

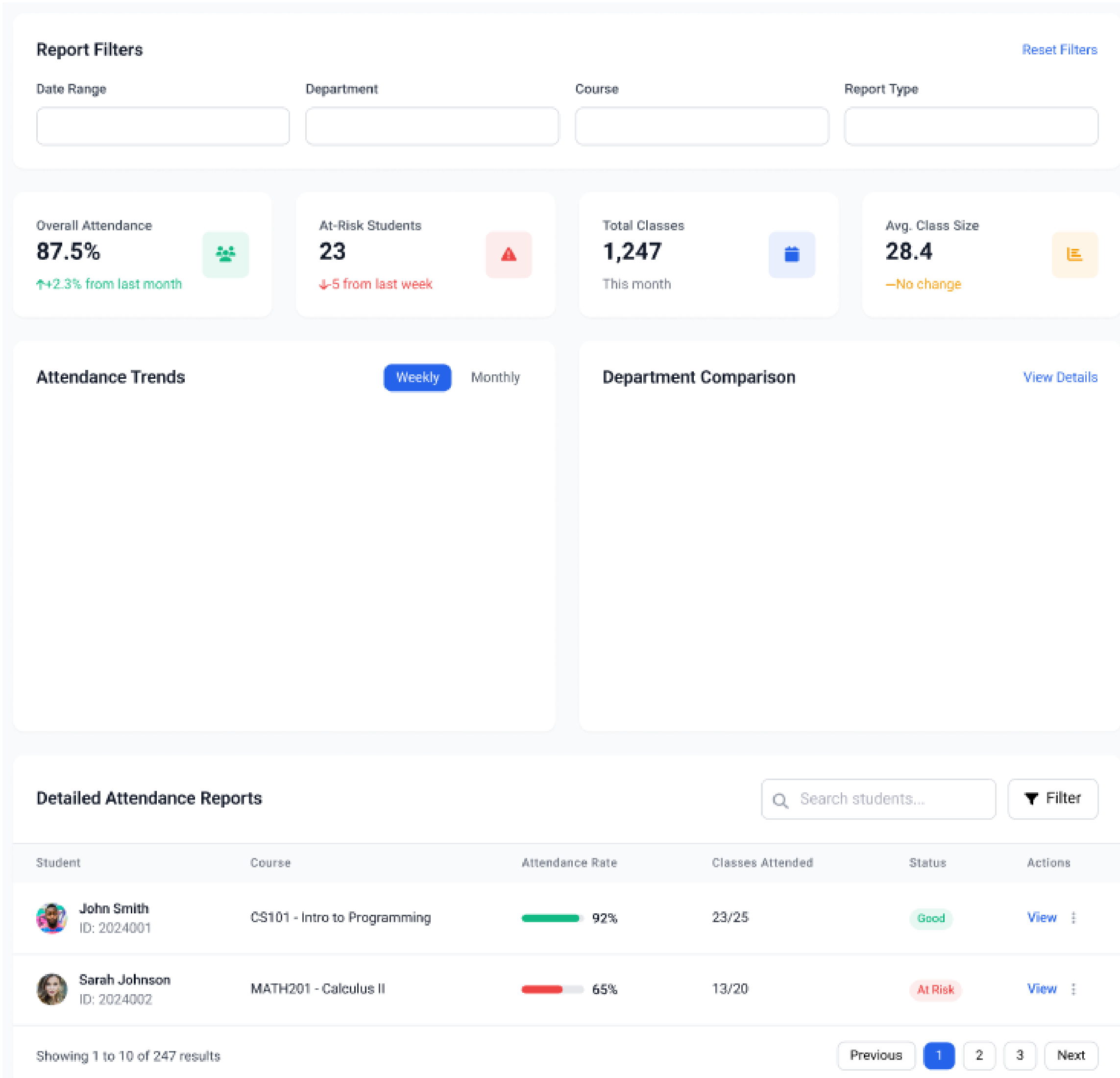

Figure 12.5: **Reports & Analytics**

Interactive dashboards show attendance trends using graphs and charts (bar, pie, line). Reports can be filtered by student, class, or date. Export buttons allow downloads in formats like PDF or CSV. Notifications highlight anomalies like frequent absences or class trends.

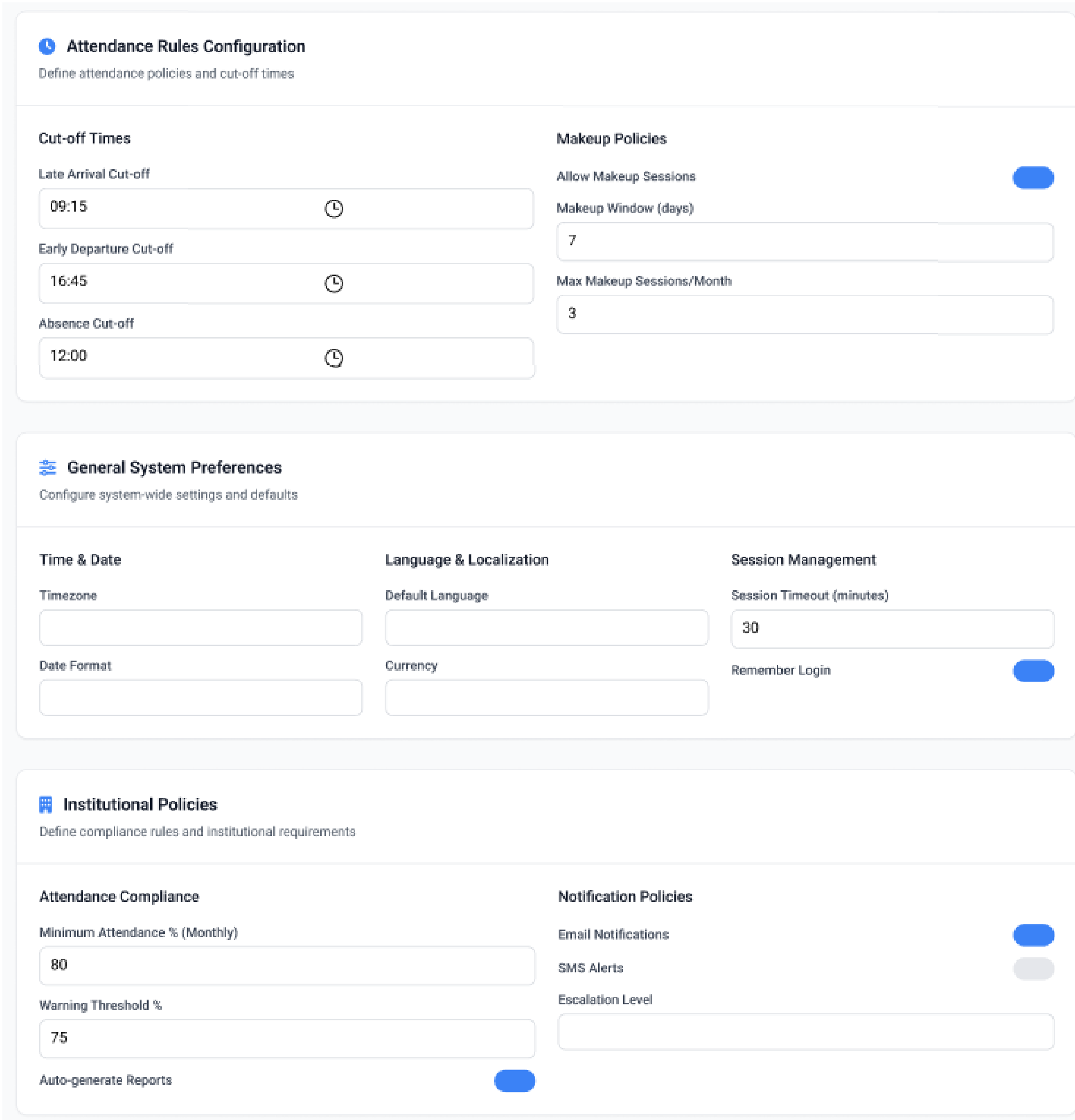

Figure 12.6: **System Configuration**

A settings panel where admins configure rules like cut-off times, makeup attendance policies, and compliance thresholds. Toggle switches, dropdowns, and text fields enable customization. A preview or simulation panel helps test changes before saving.

Database Design

Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) presents the structure of the FaceCheck system's database by showing how entities relate to one another. This diagram guides the development team in designing a clear and organized database. It also ensures that the system stores and manages data effectively, supporting the goals of accurate and automated attendance tracking.

At the heart of the system is the User entity, which stores the basic information of every person using the system—whether they are students, faculty members, or administrators. Every user is assigned a distinct role, which determines their access to system functions and is also associated with a specific department, such as a college or office.

For classroom attendance, the system includes entities like Student, Course, Class, and Faculty. Students are enrolled in courses and assigned to classes, which are scheduled and handled by faculty members. The system records attendance through the Student_Class and Attendance tables, with additional tables, such as Days and Class_Days, used to track the days on which each class is held.

The system is also designed to handle event attendance. Through the Event and Event_Attendance entities, the system can manage various types of school or organizational events, including meetings, orientations, and seminars. Users can be tagged as attendees or facilitators of events, and their participation is tracked accordingly.

Thus, the ERD reflects the system's goal of providing a flexible, reliable, and user-friendly method for managing attendance in various contexts. It ensures that all types of users are accounted for and that data is organized in a way that supports smooth operations and future scalability.

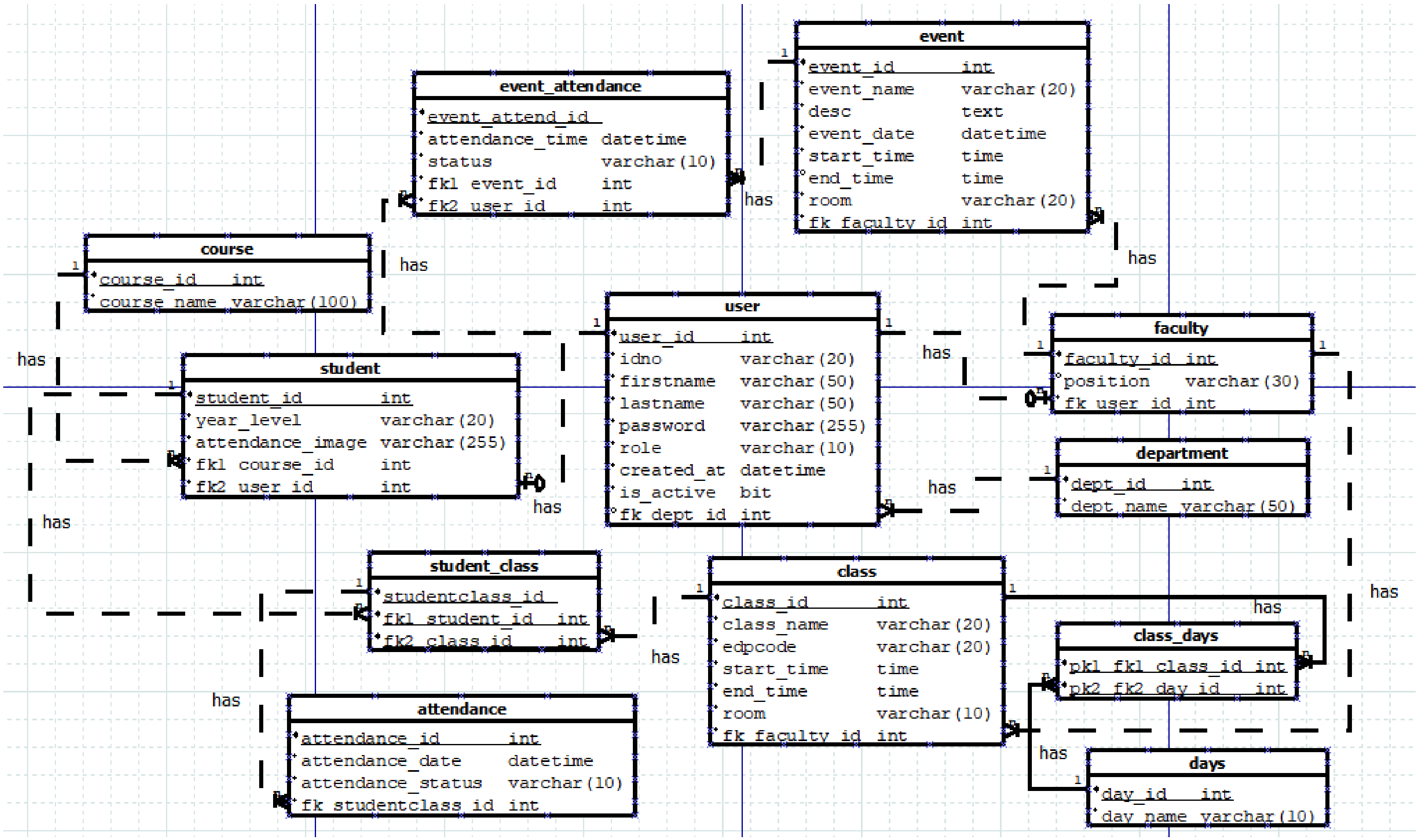

Figure 13: **Entity Relationship Diagram**

Data Dictionary

The FaceCheck Data Dictionary provides an overview of the database's structure and attributes. It guides the development team in organizing, accessing, and managing data within the application. By specifying each table and its components, the dictionary helps ensure consistent implementation and efficient system performance.

The FaceCheck system utilizes twelve core tables that represent key components, including user, role, student, faculty, course, department, class, event, and attendance. Each table includes attributes with defined data types, valid ranges, and specific constraints. The dictionary also identifies all primary and foreign keys to maintain clear relationships across tables.

The system uses consistent integer-based primary keys ranging from 1 to 999,999. It assigns varchar types for text fields, datetime and time for temporal values, and bit for binary status flags. Foreign keys maintain referential integrity and reliably connect related entities.

The FaceCheck Data Dictionary ensures that data is organized, accurate, and accessible. It provides a clear foundation for database design, development, and maintenance, enabling efficient handling of student attendance records.

ATTRIBUTE NAME	CONTENTS	TYPE	FORMAT	RANGE	REQUIRED	PK OR FK	FK REFERENCED TABLE
user_id	user id	int	999999	1-999999	y	pk	
idno	user id number	varchar(20)	xxxxx		y		
firstname	user first name	varchar(50)	xxxxx		y		
lastname	user last name	varchar(50)	xxxxx		y		
role	role of the user	varchar(10)	xxxxx		y		
password	user password	varchar(255)	xxxxx		y		
created_at	user created	datetime	yyyy-mm-dd hr:mm		y		
is_active	status of the user	bit	0,1		y		
dept_id	department id	int	999999	1-999999	y	fk	department
student_id	student id	int	999999	1-999999	y	pk	
year_level	student year level	varchar(20)	xxxxx		y		
attendance_image	student registered image	varchar(255)	xxxxx		y		
course_id	course id	int	999999	1-999999	y	fk1	course
user_id	user id	int	999999	1-999999	y	fk2	user
faculty_id	faculty id	int	999999	1-999999	y	pk	
position	faculty position	varchar(30)	xxxxx		y		
user_id	user id	int	99999	1-999999	y	fk	user
course_id	course id	int	999999	1-999999	y	pk	
course_name	course name	varchar(100)	xxxxx		y		
dept_id	department id	int	999999	1-999999	y	pk	
dept_name	department name	varchar(50)	xxxxx		y		
class_id	class id	int	99999	1-999999	y	pk	
class_name	name of the class	varchar(20)	xxxxx		y		
edpcode	educational development plan code	varchar(20)	xxxxx		y		
start_time	class start time	time	hr:mm		y		
end_time	class end time	time	hr:mm		y		
room	room of the class	varchar(10)	xxxxx		y		
faculty_id	faculty id	int	999999	1-999999	y	fk	faculty
day_id	day id	int	999999	1-999999	y	pk	
day_name	name of the day	varchar(10)	xxxxx		y		

class_id	class id	int	999999	1-999999	y	pk		
day_id	day id	int	999999	1-999999	y			
studentclass_id	student class id	int	999999	1-999999	y	pk		
student_id	student id	int	999999	1-999999	y	fk1	student
class_id	class id	int	999999	1-999999	y	fk2	class
attendance_id	attendance id	int	999999	1-999999	y	pk	
attendance_date	attendance date and time	datetime	yyyy-mm-dd hr:mm		y		
attendance_status	attendance status	varchar(10)	xxxxx		y		
studentclass_id	student class id	int	999999	1-999999	y	fk	student_class
event_id	event id	int	999999	1-999999	y	pk	
event_name	name of the event	varchar(20)	xxxxx		y		
desc	description of the event	text	xxxxx		y		
event_date	event date	datetime	yyyy-mm-dd hr:mm		y		
start_time	event start time	time	hr:mm		y		
end_time	event end time	time	hr:mm		y		
room	room of the event	varchar(20)	xxxxx		y		
faculty_id	faculty id	int	999999	1-999999	y	fk	faculty
event_attend_id	event attendance id	int	999999	1-999999	y	pk	
attendance_time	event attendance time log	datetime	yyyy-mm-dd hr:mm		y		
status	attendance status	varchar(10)	xxxxx		y		
event_id	event id	int	999999	1-999999	y	fk1	event
user_id	user id	int	999999	1-999999	y	fk2	user

Figure 14: **Database Dictionary**

Network Design

This section involves planning and structuring a network's framework to ensure it effectively supports the operations and requirements of an application or system. It involves careful consideration of the network's layout, architecture, and components to ensure optimal performance, reliability, and security.

Network Model

The FaceCheck system follows a three-tier architecture to manage attendance using facial recognition. The architecture separates the system into three functional layers: presentation, business logic, and data management.

In the presentation tier, faculty members and students use desktop or laptop computers connected to a local area network (LAN) to access the system. Faculty members monitor attendance records and manage class schedules. Students view their attendance records based on the classes for which they are enrolled. The business logic tier runs on a web server that receives user requests, processes facial recognition data, and applies attendance rules. This tier controls system behavior and manages communication between the interface and the database: the data management tier stores attendance logs, user information, and class schedules in a database. The database and web server exchange information to retrieve and update records as needed. Each tier handles a distinct set of tasks, which keeps the system organized and efficient.

Network Topology

The FaceCheck system operates on a centralized three-tier network topology within a local area network. Users connect through desktop or laptop computers in the same network. Client devices in the presentation tier send requests to the web server in the business logic tier. The web server executes system functions and exchanges data with the database server in the data management tier. The local network setup enables the system to provide fast and secure access to attendance features, eliminating the need for external internet connections.

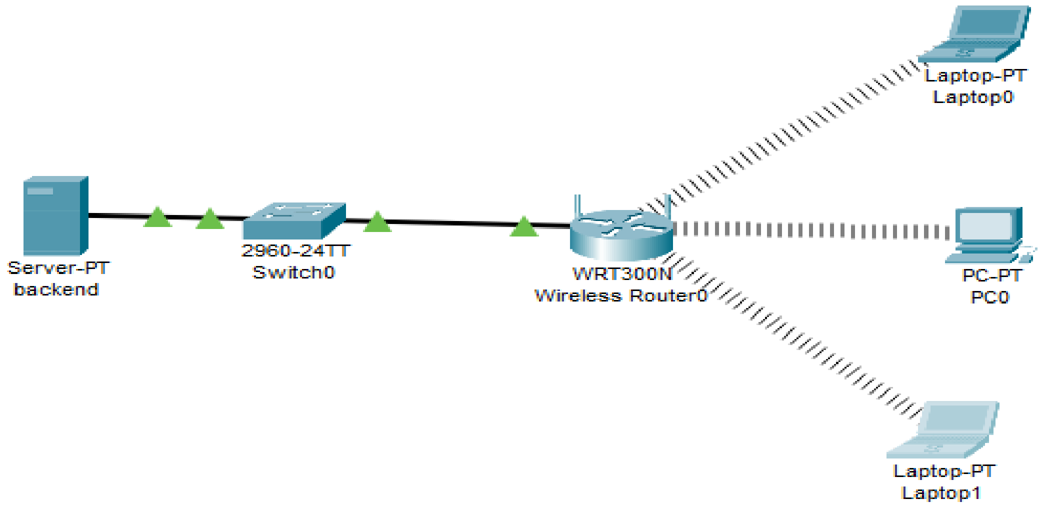

Figure 15: **Network Topology**

Development/Construction/Build Phase

This phase involves transforming designs, plans, and specifications into tangible products. During this time, several well-coordinated actions and procedures are used to bring the conceptual ideas and requirements to implement in the project.

Technology Stack

The FaceCheck system uses a structured, layered technology stack that supports its web-based facial recognition attendance features. The stack follows a modular approach, dividing responsibilities across distinct layers to promote maintainability, scalability, and efficient development.

At the top layer, the system uses Jinja2, a templating engine that integrates dynamic content into HTML pages. This component handles server-side rendering, allowing the system to inject data directly into the user interface during page generation.

The second layer splits into two components: HTML and CSS/JS. HTML provides the structural foundation of the web pages, while CSS and JavaScript manage presentation and interactivity. Together, these technologies form the presentation layer, enabling users to interact with the system via desktop or laptop computers connected to a local area network.

The third layer features OpenCV, a powerful open-source computer vision library. OpenCV handles facial recognition tasks by processing image data and comparing facial features against stored records. It plays a central role in automating attendance tracking with accuracy and efficiency.

The bottom layer represents the backend and data management functions. On the left, Flask and Python form the system's core backend framework. Flask manages routing, request handling, and web server integration, while Python provides the programming logic for operations, including facial recognition workflows and attendance validation.

On the right side of the bottom layer, SQLite serves as the database engine. It stores user information, attendance records, class schedules, and system configurations. The system uses SQLite for its simplicity and compatibility with local deployments.

This structured technology stack enables the FaceCheck system to function reliably in a local network environment, integrating real-time facial recognition with efficient data handling and a responsive user interface.

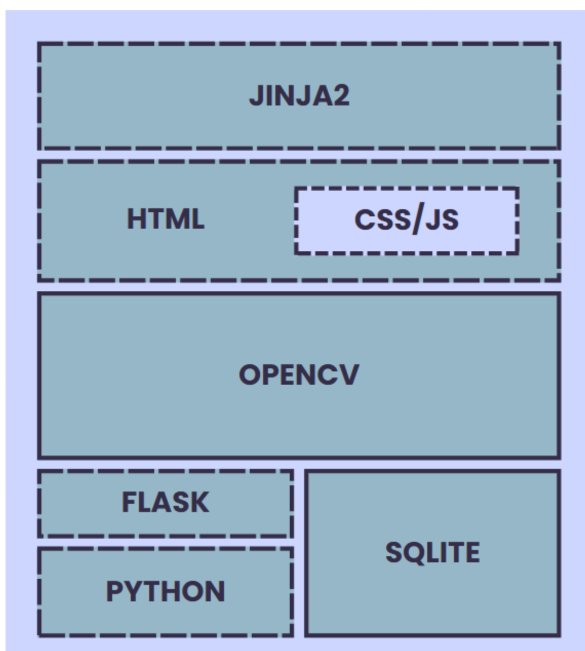

Figure 16: **Technology Stack**

Software Specification

The FaceCheck attendance monitoring system will use the following minimum specifications for its development:

Table 6

SOFTWARE SPECIFICATION

Operating System	Windows 10/11
Development Environment	Visual Studio Code
Backend Development	Python, Flask, OpenCV
Frontend Development	HTML5, CSS3, Jinja2, Javascript
UI/UX Design	Figma, Canva
Database	SQLite
Version Control	Git, Github
Testing Environment	Web Browsers, Local Server

Hardware Specification

The hardware specifications are a crucial component in ensuring the performance, accuracy, and stability of the FaceCheck attendance system. Since the project involves real-time facial recognition, data processing, and secure record-keeping, the development team has selected components that support high-resolution video input, robust computation, and secure storage. These specifications ensure the system's smooth operation under varying environmental conditions, including low-light classrooms and areas with limited internet connectivity.

The chosen hardware enables both development testing and real-world simulation, providing a foundation for evaluating recognition accuracy, latency, and multi-user throughput.

The goal is to build a cost-effective yet robust setup that can be deployed in educational institutions using standard desktops or laptops without requiring expensive or specialized equipment.

Table 7

HARDWARE SPECIFICATION

Processor (CPU)	Intel Core i3 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB (Recommended: 16 GB)
Storage	256 GB SSD (Recommended: 512 GB SSD)
Camera	HD Webcam (720p or higher)
Display	14" or larger monitor, 1080p resolution
Peripherals	Basic USB keyboard and mouse

Program Specification

Program Specification is a formal and structured description of the software's expected functionalities and the system's operational behavior. For the FaceCheck system, this specification defines the modules that must be implemented to fulfill the research objectives including automated attendance logging, facial recognition, real-time reporting, and system administration.

The FaceCheck system aims to provide secure, efficient, and hygienic attendance monitoring in educational institutions. The software is designed to run on laptops or desktops equipped with webcams and internet connectivity (for full functionality). However, the system also includes offline operation capabilities with local caching and automatic synchronization when reconnected.

List of Modules

The following list of modules details the core functionalities of the FaceCheck system, along with the assigned developers and user access levels. These modules ensure streamlined attendance tracking and system supervision across three major user groups: Students, Faculty, and System Administrators.

Table 8
LIST OF MODULES

Programmer	Modules	Attendee	Facilitator	Admin
Durano, Arcana, Ocliasa, Real	User Management Module			
	1. Create User Account			*
	2. Edit User Account			*
	3. Assign User Roles			*
	4. Reset Password		*	*
	5. Deactivate Account			*
No. of Points (*1 point per module per user*)		0	1	1
Durano, Arcana, Ocliasa, Real	Facial Registration & Recognition Module			
	1. Face Enrollment	*		
	2. Capture Face Template	*		
	3. Real-Time Face Detection	*		
	4. Liveness Detection (Anti-Spoofing)	*		

	5. Log Recognition Event	*		
No. of Points (*1 point per module per user*)		1	0	0
Durano, Arcana, Ocliasa, Real	Attendance Tracking Module			
	1. Auto Timestamp Attendance	*		
	2. Log Attendance to Database	*		
	3. View Attendance Record	*	*	*
	4. Verify Identity on Recognition	*		
	5. Prevent Duplicate Logging	*		
No. of Points (*1 point per module per user*)		1	1	1
Durano, Arcana, Ocliasa, Real	Attendance Monitoring Module			
	1. View Daily Logs	*	*	*
	2. Generate Attendance Reports		*	*
	3. Filter by Class/Date		*	*
	4. Edit/Override Attendance		*	*
	5. Export Attendance File (PDF/Excel)		*	*
No. of Points (*1 point per module per user*)		1	1	1
Durano, Arcana, Ocliasa, Real	Class & Event Management Module			
	1. Create Class & Event Schedule			*
	2. Assign Faculty to Classes & Events			*
	3. Enroll Students to Classes			*
	4. View Class & Event Lists		*	*
	5. Update Class & Event Info			*

No. of Points (*1 point per module per user*)		0	1	1
Durano, Arcana, Ocliasa, Real	Reports & Analytics Module			
	1. Create Class Schedule			*
	2. Absence Patterns Analytics		*	*
	3. Monthly Attendance Graphs		*	*
	4. Export Report CSV		*	*
	5. Compare Attendance Across Classes			*
No. of Points (*1 point per module per user*)		0	1	1
Durano, Arcana, Ocliasa, Real	System Configuration Module			5
	1. Set Attendance Rules			*
	2. Configure Email Notifications			*
	3. Adjust Recognition Sensitivity			*
	4. Change System Settings			*
	5. Manage Database Settings			*
No. of Points (*1 point per module per user*)		0	0	1
Total Number of Modules		3	5	6

Testing Plan

Unit Testing

Unit testing involves assessing individual components of the system to ensure each one functions correctly in isolation. It helps verify that each module performs its intended operations accurately.

Table 9

UNIT TESTING

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Result	Actual Result	Remarks
User Management	Create User Account		UM001	Verify that admin can create a new user with valid input.	User created successfully.		
User Management	Assign User Role		UM002	Test assigning Faculty role to a user.	Role updated to Faculty.		
User Management	Reset Password		UM003	Verify user password can be reset by admin.	Password reset email sent.		
Facial Registration & Recognition	Face Enrollment		FRR001	Test student's face registration process.	Face data saved and linked to student.		
Facial Registration & Recognition	Real-Time Face Detection		FRR002	Test face detection during attendance logging.	Face detected with bounding box.		
Facial Registration & Recognition	Liveness Detection		FRR003	Verify system prevents spoofing using static image.	Spoof attempt rejected.		
Attendance Tracking	Log Attendance to DB		AT001	Test if attendance is logged correctly.	Attendance record saved with timestamp.		

Attendance Tracking	Prevent Duplicate Logging		AT002	Attempt logging attendance twice in the same session.	Second attempt blocked.		
Attendance Monitoring	View Daily Logs		AM001	Faculty views daily logs for class.	Logs displayed correctly.		
Attendance Monitoring	Export Attendance		AM002	Export attendance logs to Excel.	File exported and downloaded.		
Class Management	Enroll Students		CM001	Admin enrolls student into a class.	Student appears in class list.		
Class Management	Assign Faculty		CM002	Assign a faculty member to a class.	Faculty successfully linked to class.		
Reports & Analytics	View Monthly Attendance Graph		RA001	Faculty views graph of monthly attendance.	Graph shows accurate trends.		
Reports & Analytics	Export Report CSV		RA002	Export attendance report to CSV.	File generated and accurate.		
System Configuration	Set Attendance Rules		SC001	Admin updates cut-off time rule.	New cut-off rule saved and active.		
System Configuration	Adjust Recognition Sensitivity		SC002	Modify sensitivity to improve accuracy.	Changes applied to recognition engine.		

Integration Testing

Integration testing examines how integrated components work together to identify interaction issues. It ensures that the interfaces between units' function correctly when combined.

Table 10
INTEGRATION TESTING

Module Name	Unit Name	Date Tested	Test Case ID	Test Case Description	Expected Results	Actual Result	Remarks
User Management + Login	User Role Access Control		IT001	Ensure student cannot access admin panel.	Access denied with proper redirection.		
Facial Registration + DB Logging	Face Enrollment Integration		IT002	After student enrolls, data should be linked and stored in the database.	Student face data saved with correct ID.		
Facial Recognition + Attendance	Attendance Logging via Recognition		IT003	After recognition, attendance must log in real-time.	Record created in the attendance table with timestamp.		
Attendance Tracking + Monitoring	View Attendance Log		IT004	Faculty should view updated logs after student logs attendance.	Real-time log appears in faculty dashboard.		
Class Management + Attendance	Filter Logs by Class/Student		IT005	Filter logs by specific class ID to ensure correct class linkage.	Logs filtered and matched to enrolled students only.		
Attendance + Reports	Generate Attendance Report		IT006	Ensure generated reports include all attendance entries.	Report contains accurate attendance records per student.		

Configuration + Recognition	Apply Cut-off Rules to Logging	2025-06-08	IT007	Set cut-off time; late students should be marked accordingly during recognition.	Late log classified correctly in the report.		
User Management + Class Mgmt	Assigned Faculty Can Only Access Own Class Logs	2025-06-08	IT008	Ensure faculty can't view attendance for unassigned classes.	Unauthorized access blocked; logs limited to assigned classes.		
Report Export + Attendance Logs	Export CSV of Filtered Results		IT009	Export class attendance report to CSV format.	CSV file downloads with correct format and values.		
System Settings + Recognition	Adjust Sensitivity & Test Detection		IT010	Change sensitivity settings and test if detection is still accurate.	Detection adapts as per new sensitivity settings.		

Alpha Testing

Alpha testing is conducted by developers or a testing team in a controlled environment to identify bugs and issues. This phase helps ensure the system meets requirements before being released to a broader audience.

Table 11
ALPHA TESTING

Test Criteria	Poor	Fair	Good	Very Good
Performance (The system responds quickly and operates smoothly under expected load conditions.)				
Security (The system ensures user data is protected and unauthorized access is prevented.)				
Compatibility (The system works well across different devices and browsers.)				
Error Handling (The system provides clear and helpful error messages.)				
Navigation (The system provides intuitive and easy-to-use navigation throughout.)				
Reversibility (The system shows clear messages or confirmation prompts that let users cancel or undo important actions, helping them correct mistakes or unintended actions.)				
Graphical User Interface (GUI)				
Simplicity (The GUI design includes simple GUI buttons, such as simple screens with clear, uncrowded messages.)				

Reusability (The system contains reusable GUI components such as familiar buttons, text and checkboxes, and other tools.)				
Consistency (The user interface maintains a cohesive and uniform design, with consistent formatting, layout, and icons throughout the system.)				
Usability (The system is user-friendly and easy to learn for new users.)				
Scalability (The system can handle increased load or usage without compromising performance.)				
Overall Impression (The general perception of the system's quality, ease of use, and value based on user experience.)				

Acceptance Testing

Acceptance testing is performed by end users or clients to validate the system against business requirements. This ensures that the system meets predefined acceptance criteria and is ready for deployment.

Table 12
ACCEPTANCE TESTING

PROJECT IDENTIFICATION	
Project Name	Date Created
FaceCheck, an AI-powered facial recognition system	June 2025

<table>
<tr><td colspan="2">Project Sponsor</td><td colspan="5">Project Manager</td></tr>
<tr><td colspan="2">Durano, Arcana, Ocliasa, Real</td><td colspan="5">Jhanna Kris Durano</td></tr>
<tr><td colspan="2">Project Adviser</td><td colspan="5">Dean</td></tr>
<tr><td colspan="2">Dennis S. Durano</td><td colspan="5">Mr. Neil A. Basabe</td></tr>
<tr><td colspan="7">ACCEPTANCE TEST CRITERIA MATRIX</td></tr>
<tr><td colspan="2"></td><td colspan="2">Critical</td><td colspan="2">Test Results</td><td></td></tr>
<tr><td>Number</td><td>Acceptance Requirements</td><td>Yes</td><td>No</td><td>Accept</td><td>Reject</td><td>Comments</td></tr>
<tr><td>1</td><td>The system shall allow students to register their face for attendance.</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>2</td><td>The system shall correctly recognize a registered student’s face and log their attendance with at least 95% accuracy.</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>3</td><td>The system shall reject unregistered or invalid faces.</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>4</td><td>The facial recognition process must complete within 3 seconds.</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>5</td><td>The system shall log a student's attendance automatically upon successful facial recognition.</td><td></td><td></td><td></td><td></td><td></td></tr>
<tr><td>6</td><td>The system shall allow students to view their own attendance records securely.</td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

7	The system shall prevent students from logging attendance outside their scheduled class time.					
8	Faculty shall be able to view the attendance records of students in their assigned classes.					
9	Faculty shall be able to generate class-based attendance reports.					
10	Faculty shall be able to override or update attendance entries for valid reasons (e.g., incorrect auto-log, class dismissal).					
11	The system shall allow administrators to add, edit, or remove user accounts (students and faculty).					
12	The system shall allow administrators to manage classes and assign faculty and students.					
13	The system shall support the generation of institution-level attendance reports.					
14	The system shall include basic configuration settings for system behavior and access levels.					
15	Users shall only access the features appropriate to their role.					
16	The system shall securely handle facial data and login credentials.					
17	Unauthorized access to facial recognition features or data should be prevented.					
18	The system shall operate consistently under normal load					

	conditions (e.g., simultaneous log-ins).					
19	Facial recognition should work with minor appearance changes (e.g., glasses, haircut).					
20	Error messages and prompts must guide users effectively in case of failure.					
OTHER CONSIDERATIONS						
Business Objectives - Did the web application/system meet the business objective?						
Yes	No					
Does the web application/system require any setups prior to installation? If so please describe.						
Yes	No					
APPROVAL						
Respondent		Signature			Date	
Yes	No					

CHAPTER IV

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

Data Presentation

The data collected through the questionnaire was analyzed to evaluate the effectiveness of the current attendance monitoring system, expectations for FaceCheck, and its potential impact. Responses were categorized into sections based on demographics, current attendance experience, familiarity with facial recognition, data privacy and security considerations, system usability and feature preferences, and overall feedback for the new system.

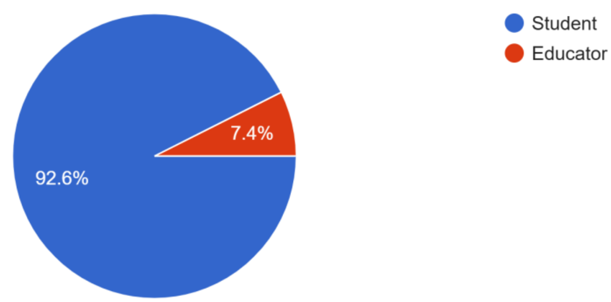

Figure 17: **Pie Chart**

A pie chart that presents the demographic distribution of respondents, categorized as student and educator.

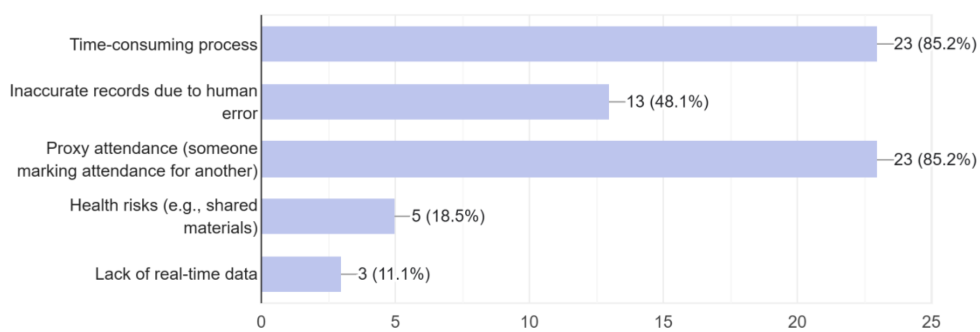

Figure 18: **Bar Graph 1**

A bar graph that highlights common challenges faced with the existing system, such as time-consuming process, inaccurate to records, and proxy attendance.

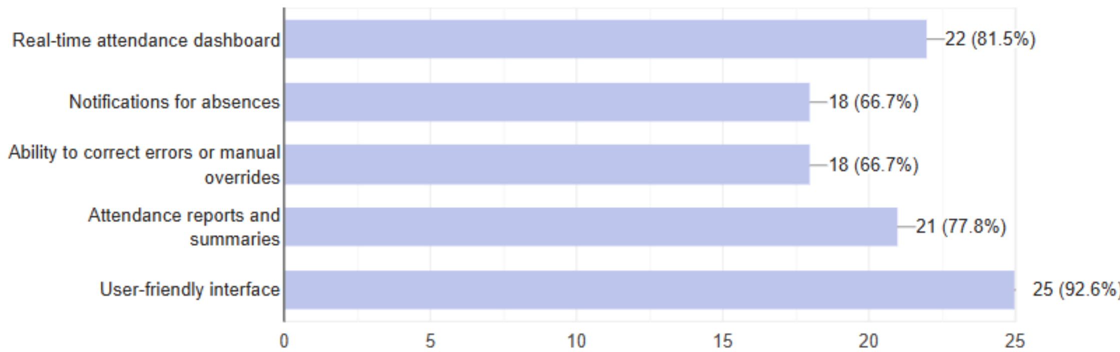

Figure 19: **Bar Graph 2**

A line chart that tracks the perceived importance of new features, including user-friendly interfaces, real-time attendance dashboard, and attendance reports and summaries.

Data Analysis

Demographics

The majority of respondents were students (92.6%), with educators comprising a smaller but significant portion (7.4%). This distribution indicates that the survey primarily captured student perspectives, making up 15% of the total sample. Most respondents represent the primary user base of the institution, indicating comprehensive feedback from the main stakeholder group.

Academic Level Distribution (Students Only):

Based on the 25 student responses, the year level breakdown reveals:

- 59.3% of student respondents are in their 3rd year, representing the largest segment of participants
- 29.6% are 4th-year students, indicating strong participation from senior students
- 7.4% are 2nd-year students, showing moderate representation from intermediate levels
- 3.7% are 1st year students, with minimal participation from newcomers

Current Attendance System Evaluation

Current Attendance Methods:

- Manual roll call remains the dominant method at 26 responses (96.3%), indicating heavy reliance on traditional attendance taking

- Paper log sheets are used by 18 respondents (66.7%), showing continued dependence on physical documentation

- Digital systems (e.g., online portals) are utilized by 14 respondents (51.9%), demonstrating growing adoption of technology-based solutions

Common Challenges:

- Time-consuming process was cited by 23 respondents (85.2%), representing the most critical operational challenge

- Proxy attendance (someone marking attendance for another) affects 23 respondents (85.2%), indicating serious integrity concerns

- Inaccurate records due to human error impact 13 respondents (48.1%), highlighting reliability issues with manual systems

- Health risks (e.g., shared materials) concern 5 respondents (18.5%), reflecting awareness of hygiene considerations

- Lack of real-time data affects 3 respondents (11.1%), showing limited demand for immediate attendance information

Data Interpretation

The findings provide a comprehensive understanding of the institution's current attendance system and expectations for modernization:

1. System Inefficiencies:

Time-consuming processes, integrity issues, and limited technological integration hinder the current attendance system. The overwhelming majority of respondents (81.5%) engage with attendance monitoring daily, yet 85.2% cite time-consuming processes as a major challenge. This suggests that despite regular usage, the system has a significant impact on operational efficiency.

2. Demand for Digital Solutions:

Respondents' mixed usage of attendance methods demonstrates a transition period where traditional approaches (manual roll call at 96.3%) coexist with emerging digital solutions (51.9%). The high adoption rate of manual methods alongside growing digital system usage suggests readiness for comprehensive technological integration.

3. Academic Integrity Concerns:

While many respondents are familiar with current attendance processes, proxy attendance issues affect 85.2% of users, highlighting the critical need for verification mechanisms. The equal prevalence of time-consumption and integrity issues underscores the dual challenge facing institutions.

4. Implementation Readiness:

The demographic composition, with 88.9% of responses from upper-level students (3rd and 4th year), provides experienced user perspectives essential for system development. The predominant student participation (92.6%) ensures end-user requirements are well-represented, though limited educator input (7.4%) suggests a need for additional stakeholder engagement.

Conclusion

The feedback confirms the need for a modernized attendance system as a solution to the inefficiencies of the current institutional management approach. The findings demonstrate its potential to:

1. Streamline attendance recording processes through automated solutions.
2. Enhance academic integrity using digital verification tools.
3. Provide efficient, real-time access to attendance data.

By addressing respondents' concerns about time efficiency, accuracy, and system reliability, a modernized attendance system can significantly improve institutional operations and user satisfaction. The high engagement levels and clear identification of system limitations indicate strong institutional readiness for technological advancement in attendance management.

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CURRICULUM VITAE

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Address	: Sambag-4, Guadalupe, Cebu City
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Name of Mother	: Nenalyn G. Landao

EDUCATIONAL ATTAINMENT

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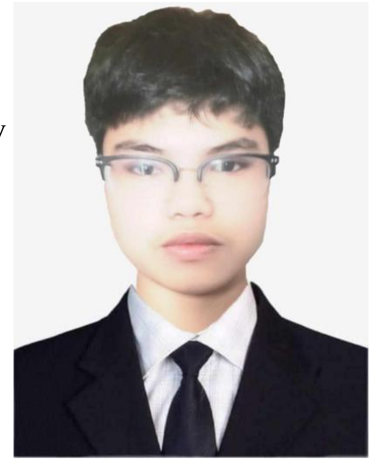
Junior High School	: Naga National High School 878 Bonifacio St, City of Naga, Cebu 2019 – 2020
Senior High School	: Bantayan National High - Senior High School Accountancy, Business, and Management (ABM) Ticad, Bantayan, Cebu 2021 – 2022
College	: University of Cebu Main Campus Bachelor of Science in Information Technology (BSIT) Sanciangko St, Cebu City, 6000 Cebu 2022 - Present



PERSONAL DATA

Name : Ocliasa, Niño Rollane D.

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Date of Birth : June 8, 2004

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Citizenship : Filipino

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Name of Mother : Mylin D. Ocliasa

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2019 – 2020

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2021 – 2022

College : University of Cebu Main Campus
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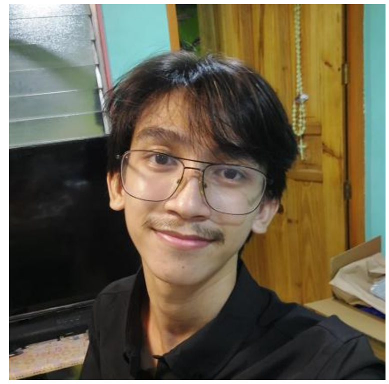
2022 - Present

PERSONAL DATA



Name : Real, Rovic Steve R

Address : 0241 Ermita Proper, Cebu City

Date of Birth : October 8, 2003

Civil Status : Single

Citizenship : Filipino

Email : rovic.steve@gmail.com

Name of Father : Esteban A. Real Jr.

Name of Mother : Evelyn R. Real

EDUCATIONAL ATTAINMENT

Elementary : City Central School
P. del Rosario St., Brgy Sambag I, Cebu City
2015 - 2016

Junior High School : Abellana National School
Osmeña Boulevard, Cebu City
2019 – 2020

Senior High School : Cebu Institute of Technology - University
Science, Technology, Engineering, and Mathematics (STEM)
N. Bacalso Avenue, Cebu City 6000, Philippines
2021 – 2022

College : University of Cebu Main Campus
Bachelor of Science in Information Technology (BSIT)
Sanciangko St, Cebu City, 6000 Cebu
2022 - Present

PERSONAL DATA

Name : Arcana, Sean Joseph C.

Address : Basak San Nicolas, Cebu City

Date of Birth : September 18, 2003

Civil Status : Single

Citizenship : Filipino

Email : seanjosepharcana@gmail.com

Name of Father : Magdalino A. Arcana

Name of Mother : Josephine C. Arcana

EDUCATIONAL ATTAINMENT

Elementary : Mambaling Elementary School
Natalio B. Bacalso Ave, Cebu City, 6000 Cebu
2015 - 2016

Junior High School : University of Cebu - Main Campus
Sanciangko St, Cebu City, 6000 Cebu
2019 – 2020

Senior High School : University of Cebu - Pri
Technical Vocational - Livelihood (TVL)
J. Alcantara, Cebu City, Cebu, Philippine
2021 – 2022

College : University of Cebu - Main Campus
Bachelor of Science in Information Technology (BSIT)
Sanciangko St, Cebu City, 6000 Cebu
2022 - Present

APPENDICES

Appendix A: Capstone40 Oath of Confirmation Form

University of Cebu
College of
Computer Studies

OATH OF CONFIRMATION

This is to attest that I, Jhama Kris Durano, a **BSIT-4** student currently enrolled in the course **Capstone41/IT-CPSTONE41 (Capstone Project 1)**, have aptly received a copy; religiously read and conscientiously understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the **Capstone Project Manual 2014** document implemented by the **College of Computer Studies**, University of Cebu (Main Campus).

This is also to affirm that said guidelines was judiciously discussed and explicitly elaborated in a forum conducted by the Faculty and Dean of the college.

CONFORME:
DURANO, JHANNA KRIS 03/31/25
(Signature of Student over printed name) / Date

CONFORME:
LEXTER P. DURANO
(Signature of Parent/Guardian over printed name)

NOTED:
DENNIS DURANO
(Signature of Adviser over printed name)

NOTED:
NEIL A. BASABE, MIT
Dean, UC Main - CCS
Dean

University of Cebu
College of
Computer Studies

OATH OF CONFIRMATION

This is to attest that I, ROVIC STEVE R. REAL, a **BSIT-4** student currently enrolled in the course **Capstone41/IT-CPSTONE41 (Capstone Project 1)**, have aptly received a copy; religiously read and conscientiously understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the **Capstone Project Manual 2014** document implemented by the **College of Computer Studies**, University of Cebu (Main Campus).

This is also to affirm that said guidelines was judiciously discussed and explicitly elaborated in a forum conducted by the Faculty and Dean of the college.

CONFORME:
REAL, ROVIC STEVE R. 03/31/25
(Signature of Student over printed name) / Date

CONFORME:
[illegible] R. Real
(Signature of Parent/Guardian over printed name)

NOTED:
(Signature of Adviser over printed name)

NOTED:
NEIL A. BASABE, MIT
Dean, UC Main - CCS
Dean

University of Cebu
College of
Computer Studies

OATH OF CONFIRMATION

This is to attest that I, Niño Rollane D. Ocliasa, a **BSIT-4** student currently enrolled in the course **Capstone41/IT-CPSTONE41 (Capstone Project 1)**, have aptly received a copy; religiously read and conscientiously understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the **Capstone Project Manual 2014** document implemented by the **College of Computer Studies**, University of Cebu (Main Campus).

This is also to affirm that said guidelines was judiciously discussed and explicitly elaborated in a forum conducted by the Faculty and Dean of the college.

CONFORME:

NIÑO ROLLANE D. OCLIASA 03/31/2025
(Signature of Student over printed name) / Date

CONFORME:

[illegible] OCLIASA
(Signature of Parent/Guardian over printed name)

NOTED:

DENNIS DURANO
(Signature of Adviser over printed name)

NOTED:

NEIL A. BASABE, MIT
Dean, UC Main - CCS
Dean

University of Cebu
College of
Computer Studies

OATH OF CONFIRMATION

This is to attest that I, Sean Joseph C. Arcana, a **BSIT-4** student currently enrolled in the course **Capstone41/IT-CPSTONE41 (Capstone Project 1)**, have aptly received a copy; religiously read and conscientiously understood its contents; and openly submit to the terms, rules, conditions and regulations stated in the **Capstone Project Manual 2014** document implemented by the **College of Computer Studies**, University of Cebu (Main Campus).

This is also to affirm that said guidelines was judiciously discussed and explicitly elaborated in a forum conducted by the Faculty and Dean of the college.

CONFORME:

03/31/25
Sean Joseph Arcana
(Signature of Student over printed name) / Date

CONFORME:

3-31-25
JOSEPHINE ARCANA
(Signature of Parent/Guardian over printed name)

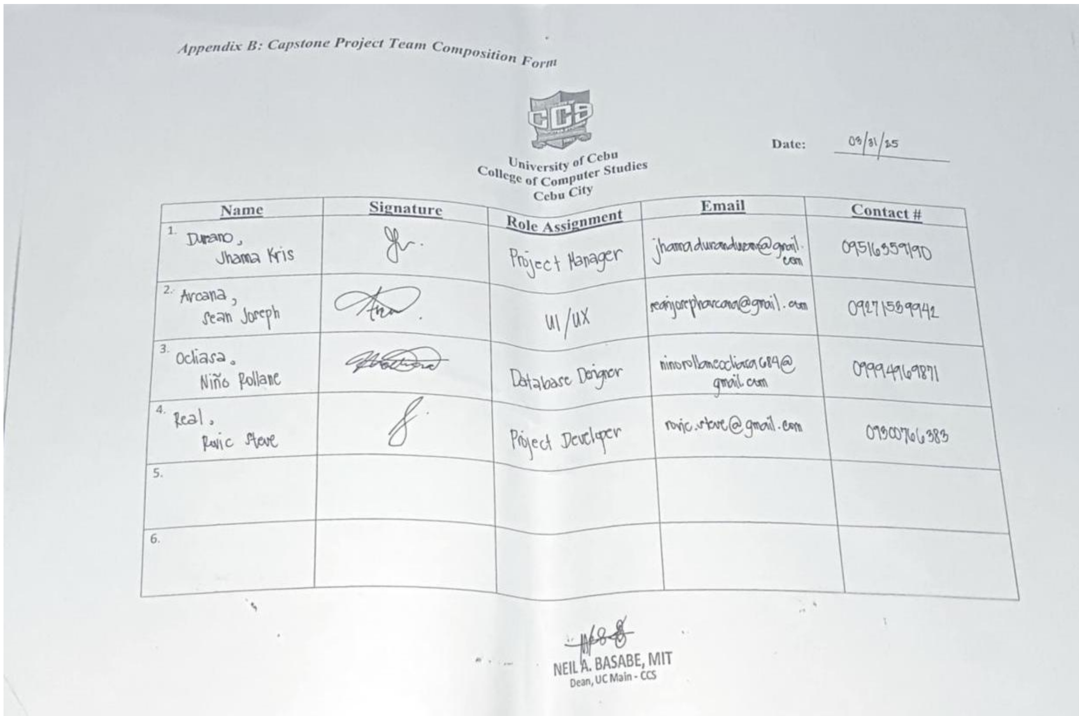
NOTED:

DENNIS DURANO
(Signature of Adviser over printed name)

NOTED:

NEIL A. BASABE, MIT
Dean, UC Main - CCS
Dean

Appendix B: Capstone Project Team Composition Form



Appendix B: Capstone Project Team Composition Form

University of Cebu
College of Computer Studies
Cebu City

Date: 03/31/25

Name	Signature	Role Assignment	Email	Contact #
1. Durano, Jhama Kris		Project Manager	jhamaduran[illegible]@gmail.com	09516359190
2. Arcana, Sean Joseph		UI/UX	seanjosepharcana@gmail.com	09271539942
3. Ocliasa. Niño Rollane		Database Designer	ninorollaneocliasa684@gmail.com	09994969871
4. Real, Rovic Steve		Project Developer	rovic.steve@gmail.com	09300766383
5.				
6.				

NEIL A. BASABE, MIT
Dean, UC Main - CCS

Appendix C: Capstone Project Business Model Canvas

University of Cebu
College of Computer Studies
Cebu City

KEY PARTNERS

- Camera Manufacturer
- Local Hardware Supplier
- Software Libraries Holder

KEY ACTIVITIES

- AI Model Development
- Facial Recognition Optimization
- Develop User Interface
- System Integration & Testing
- Market System

KEY RESOURCES

- AI Development Team
- OpenCV Libraries
- Hardware Infrastructure
- Developments Tools & Software
- Database Management

VALUE PROPOSITION

- To develop a secure & contactless attendance
- To integrate a proxy attendance prevention
- To provide real-time analytics & reporting

CUSTOMER RELATIONSHIP

- Customer Support
- System Updates
- User Training
- Hardware Facilities Recommendation
- Feedback Collection

CHANNELS

- Direct sales to schools
- Campus promotions
- Website & social media
- Training Workshops & Seminars

CUSTOMER SEGMENT

- Educational Institutions
- Corporate Businesses

COST STRUCTURE

- Research and Development Expenses
- Development and Programming Costs
- Hardware Infrastructure Costs

REVENUE STREAM

- Subscription Fees
- Training & Support Services
- Maintenance & Update Contracts

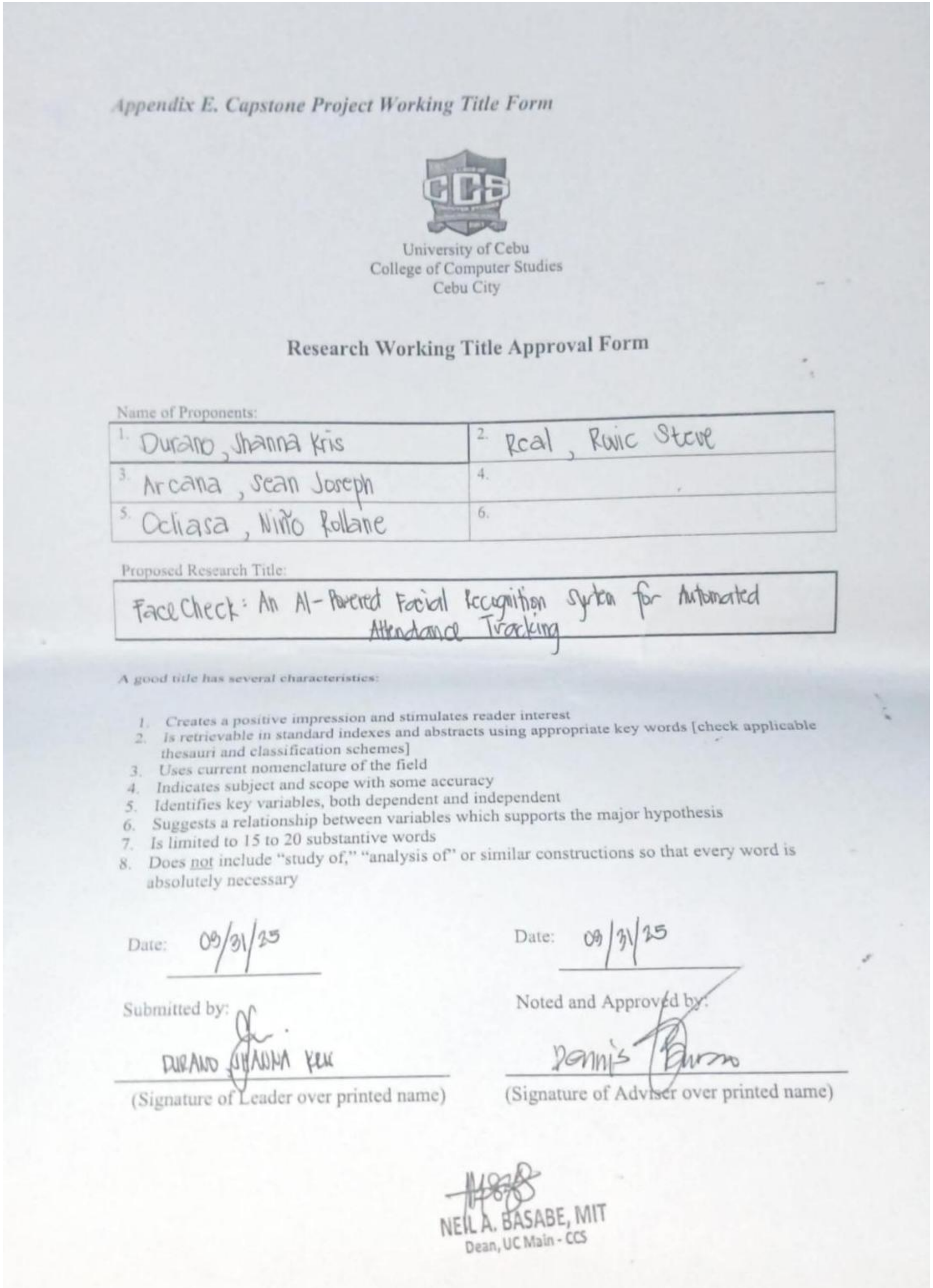
Appendix D: Capstone Project Related Studies Comparative Matrix

University of Cebu
College of Computer Studies
Cebu City

Related Studies	Features	Limitations	Platform Details	Support
Name: Sprout MySolutions URL: https://market.sprout.ph/solution/mysolutions/ Proponent(s): Sprout Solutions Philippines	1. Fingerprint, RFID & facial recognition 2. Real-time attendance logging 3. Biometric terminal with face template storage	1. Requires proprietary hardware 2. Designed for corporate use, not education 3. No support for academic schedules or reports	1. Hardware-dependent system 2. Integrated with HR and payroll platforms 3. Web-based admin tools	1. Vendor-based support 2. Technical servicing 3. System updates via vendor
Name: ZKTeco EFace10 URL: https://www.zkteco.com/en/Visiblie_Light/EFace10 Proponent(s): ZKTeco Co., Ltd.	1. Facial and fingerprint recognition 2. Touchscreen interface 3. Mask-compatible facial detection 4. Handles large attendance volumes	1. Expensive biometric hardware 2. Raw logs need external processing 3. Not suited for academic workflows	1. Biometric terminal 2. Works with ZKTeco software suite 3. On-premises deployment	1. Vendor manuals 2. Product warranty 3. Device-specific support
Name: eLeave Philippines URL: https://eleave.ph/face-recognition-attendance/ Proponent(s): E-Leave Management Systems Inc.	1. Facial recognition with IR cameras 2. Works in standalone or cloud-based mode 3. Leave and attendance integration	1. Tailored for employees, not students 2. Specialized hardware increases cost 3. No student class support	1. Infrared-enabled cameras 2. Desktop or web client 3. Integrated with HR modules	1. Email and phone support 2. Business-grade SLA 3. No education-specific helpdesk
Name: QRpho URL: https://qrpho.com/ Proponent(s): Astech Systems Corporation	1. Dynamic QR code-based check-in 2. Mobile phone scan support 3. Web admin dashboard	1. No biometric validation 2. Susceptible to proxy attendance 3. Lacks academic-	1. Web-based QR system 2. Mobile phone scanning 3. Cloud data logging	1. Limited documentation 2. Online FAQ and contact form

		specific automation		
Name: FaceCheck URL: [Internal System - University of Cebu] Proponent(s): 1. Durano, Jhanna Kris 2. Real, Rovic Steve 3. Ocliasa, Nino Rollane 4. Arcana, Sean Joseph Advisor: Dennis Durano	1. Facial recognition using webcams 2. Works on regular laptops/desktops 3. Supports class schedules and EDP codes 4. Real-time academic logging 5. Generates academic attendance reports	1. Webcam must be clear and functional 2. Requires accurate face data for training 3. Dependent on lighting and camera quality	1. Software-based system 2. Web app 3. Uses standard webcam hardware	1. In-house tech support 2. Education-specific interface 3. Easily customizable for school policies

Appendix E: Capstone Project Working Title Form



Appendix E. Capstone Project Working Title Form

University of Cebu
College of Computer Studies
Cebu City

Research Working Title Approval Form

Name of Proponents:

1. Durano, Shanna Kris	2. Real, Ruvic Steve
3. Arcana, Sean Joseph	4.
5. Ocliasa, Niño Rollane	6.

Proposed Research Title:

FaceCheck: An AI-Powered Facial Recognition System for Automated Attendance Tracking

A good title has several characteristics:

1. Creates a positive impression and stimulates reader interest
2. Is retrievable in standard indexes and abstracts using appropriate key words [check applicable thesauri and classification schemes]
3. Uses current nomenclature of the field
4. Indicates subject and scope with some accuracy
5. Identifies key variables, both dependent and independent
6. Suggests a relationship between variables which supports the major hypothesis
7. Is limited to 15 to 20 substantive words
8. Does not include "study of," "analysis of" or similar constructions so that every word is absolutely necessary

Date: 08/31/25

Submitted by:

DURANO SHANNA KRIS

(Signature of Leader over printed name)

Date: 08/31/25

Noted and Approved by:

Dennis Durano

(Signature of Adviser over printed name)

NEIL A. BASABE, MIT
Dean, UC Main - CCS

Appendix F: Capstone Project Gantt Chart

University of Cebu
College of Computer Studies
Cebu City

Iterative Gantt Chart: FaceCheck

	Planned
	Actual

Features	PLANNED ITERATION START	PLANNED ITERATIONS	ACTUAL ITERATION START	ACTUAL INTERATION	TASK COMPLETED	MAY				JUNE				JULY				AUGUST				SEPTEMBER				OCTOBER				NOVEMBER			
						1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
CHAPTER I																																	
Rationale	3	4	3	4	100%																												
Objectives	3	4	3	4	100%																												
Scope and Limitations	3	4	3	4	100%																												
Significance	3	4	3	4	100%																												
Definition of Terms	3	4	3	4	100%																												
CHAPTER II																																	
Related Literature	3	4	3	4	100%																												
Related Studies	3	4	3	4	100%																												

Comparative Matrix	3	4	3	4	100%
CHAPTER III					
Research Design					
Method	2	3	2	3	100%
Flow of the Study	2	3	2	3	100%
Research Environment	2	3	2	3	100%
Respondents	2	3	2	3	100%
Research Instrument	2	3	2	3	100%
Research Procedure					
Treatment of Data	2	3	2	3	100%
Ethical Considerations	2	3	2	3	100%
Software Engineering Methodology	2	3	2	3	100%
Planning/Conception-Initiation Phase					
Business Model Canva	2	3	2	3	100%
Program Workflow	2	3	2	3	100%
Validation Board	2	3	2	3	100%
Business Roadmap	2	3	2	3	100%
Functional Decomposition Diagram	2	3	2	3	100%
Gantt Chart	2	3	2	3	100%
Analysis-Design Phase					
Use Case Diagram	2	3	2	3	100%

Story Board	2	3	2	3	100%
Database Design	2	3	2	3	100%
Entity-Relationship	2	3	2	3	100%
Data Dictionary	2	3	2	3	100%
Network Design					
Network Model	2	3	2	3	100%
Network Topology	2	3	2	3	100%
Development/Construction Build Phase					
Technology Stack	4	1	4	1	100%
Software Specification	4	1	4	1	100%
Hardware Specification	4	1	4	1	100%
Program Specification	4	1	4	1	100%
List of Modules	4	1	4	1	100%
Testing Plan					
Unit Testing	4	1	4	2	100%
Integration Testing	4	1	4	2	100%
Alpha Testing	4	1	4	2	100%
Acceptance Testing	4	1	4	2	100%
Curriculum Vitae	4	1	4	2	100%
Modules					
User Management Module					

Create User Account	1	2			
Edit User Account	1	2			
Assign User Roles	1	2			
Reset Password	1	2			
Deactivate Account	1	2			
Facial Registration & Recognition Module					
Face Enrollment	2	4			
Capture Face Template	2	4			
Real-Time Face Detection	2	4			
Liveness Detection (Anti-Spoofing)	2	4			
Log Recognition Event	2	4			
Attendance Tracking Module					
Auto Timestamp Attendance	4	2			
Log Attendance to Database	4	2			
View Attendance Record	4	2			
Verify Identity on Recognition	4	2			
Prevent Duplicate Logging	4	2			
Attendance Monitoring Module					
View Daily Logs	2	4			
Generate Attendance Report	2	4			
Filter by Class/Date	2	4			

Edit/Override Attendance	2	4			
Export Attendance File (PDF/Excel)	2	4			
Class & Event Management Module					
Create Class & Event Schedule	4	2			
Assign Faculty to Classes & Events	4	2			
Enroll Students to Classes	4	2			
View Class & Event Lists	4	2			
Update Class & Event Info	4	2			
Report & Analytics Module					
Create Class Schedule	2	4			
Absence Patterns Analytics	2	4			
Monthly Attendance Graphs	2	4			
Export Report CSV	2	4			
Compare Attendance Across Classes	2	4			
System Configuration Module					
Set Attendance Rules	4	2			
Configure Email Notifications	4	2			
Adjust Recognition Sensitivity	4	2			
Change System Settings	4	2			
Manage Database Settings	4	2			
Testing					

Unit Testing	4	2																															
Integration Testing	4	2																															
Alpha Testing	4	2																															
Acceptance Testing	4	2																															

Appendix G: Consultation Logs Form

University of Cebu - Main
College of Computer Studies
IT-CPSTONE30/CAPSTONE41 Consultation Logs Form

Capstone Project Title:	**FACECHECK: An AI powered Facial Recognition System for Automated Attendance Tracking**
Name of Proponents:	DURANO, Jhanna Kris ARCANA, Sean Joseph OCLIASA, Niño Rollane REAL, Rovic Steve

1st Consultation:	should be with in: Chapter 1 must be completely delivered for adviser's evaluation	
Date & Time of Consultation:	6/7/2025 11:11	Dennis Durano
Team Members Present:	DURANO, Jhanna Kris ARCANA, Sean Joseph OCLIASA, Niño Rollane REAL, Rovic Steve	
Suggestions & Recommendations:	*make a bulleted statement of the problem* *revise the word "implement" to "develop"* *create a rationalized bulleted scope and limitation*	

2nd Consultation:	should be with in: Chapters 1 and 2 must be completely delivered for adviser's evaluation	
Date & Time of Consultation:	6/7/2025 11:30	Dennis Durano
Team Members Present:	DURANO, Jhanna Kris ARCANA, Sean Joseph OCLIASA, Niño Rollane REAL, Rovic Steve	
Suggestions & Recommendations:	*okey* *remove the offline functionality in the mentioned features*	

3rd Consultation:	should be with in: Chapters 1 , 2 and 3 must be completely delivered for adviser's evaluation	
Date & Time of Consultation:	6/7/2025 12:15	Dennis Durano
Team Members Present:	✓	DURANO, Jhanna Kris
	✓	ARCANA, Sean Joseph
	✓	OCLIASA, Niño Rollane
	✓	REAL, Rovic Steve
Suggestions & Recommendations:	✓	*change development model from AGILE to ITERATIVE*
	✓	*use the appropriate GANTT chart for the aforementioned*
	✓	*development model*

4th Consultation:	should be with in: Chapters 1 , 2 , 3 including initial and final pages must be completely delivered for adviser's evaluation	
Date & Time of Consultation:	6/9/2025 12:01 PM	Dennis Durano
Team Members Present:	✓	DURANO, Jhanna Kris
	✓	ARCANA, Sean Joseph
	✓	OCLIASA, Niño Rollane
	✓	REAL, Rovic Steve
Suggestions & Recommendations:	✓	OKEY NA!
	✓	CHECK FOR TYPO-ERRORS

This is to certify that I have been regulary consulted by my advisees; have reviewed their system prototype as well as the required manuscript of the above-stated study. As their adviser, I therefore submit them ready for **Proposal Hearing** *as the Chapters 1 through 3 and the pertinent parts of their manuscript are complete.*

Signed: Dennis S. Durano
(Signature of Adviser over printed name)

Appendix L: Capstone Project Hearing Notice Form

ne Project Hearing Notice

UNIVERSITY OF CEBU
College of Computer Studies

PROJECT/ RESEARCH HEARING NOTICE

Date filed: ___________
Ref. Code: ___________

[✔] PROPOSAL
[ ] FINAL ORAL EXAMINATION

Research Hearing
Date: June 18, 2025   Time: 3:30- 4:30 PM   Venue: UC MAIN CCS 538 (CCS MECCA)

Research Title:
FACECHECK: AN AI-POWERED FACIAL RECOGNITION SYSTEM FOR AUTOMATED ATTENDANCE TRACKING

Proponents/Researchers:

DURANO, JHANNA KEN	OCHAVA, NIÑO ROLLANE	
ARCANA, SEAN JOSEPH	REAL, ROVIC STEVE	

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research. [Please **PRINT NAME** and **SIGN**]

ENGR. DENNIS S. DURANO
RESEARCH ADVISER

MR. WILSON GAYO
PANEL MEMBER 1

MR. HEUBERT M. FEROLINO
PANEL MEMBER 2/Content Expert

MR. HEUBERT M. FEROLINO
PROGRAM RESEARCH COORDINATOR

MR. NEIL A. BASABE
PANEL CHAIR

NOTED/APPROVED BY:

NEIL A. BASABE, MIT
DEAN

Appendix M: Capstone Project Panel Honoraria (Form 10)

Qualitative Research

university of cebu

Appendix R
Standard Form for Honorarium, Undergraduate Thesis
Proposal Hearing/Initial Review

UNIVERSITY OF CEBU RESEARCH CENTER

STANDARD FORM NUMBER 10
HONORARIUM FOR PANEL, UNDERGRADUATE THESIS (PROPOSAL HEARING)

Position	Honoraria	Name	Conforme
Adviser	P1,000.00	ENGR. DENNIS S. DURANO	
Chairman	P500.00	MR. NEIL A. BASABE	
Member 1	P300.00	MR. WILSON GAYO	
Member 2	P300.00	MR. HEUBERT M. FEROLINO	
REC	P500.00	Ms. Rithsun Mamacos	
Administrative Fee	P500.00		
Total:	**P3,100.00**		

Name of the Lead Researcher:	JHANNA KEI DURANO
College/Department:	COLLEGE OF COMPUTER STUDIES
Campus:	UNIVERSITY OF CEBU-MAIN
Title of the Research:	FACECHECK: AN AI-POWERED FACIAL RECOGNITION SYSTEM FOR AUTOMATED ATTENDANCE TRACKING

Prepared by:

JHANNA KEI DURANO

Name & Signature of the Researcher

Approved by:	Name	Signature
Research Instructor	MRS. JENNIFER G. AMORES	
Program Research Coordinator	MR. HEUBERT M. FEROLINO	
College Dean	MR. NEIL A. BASABE	
Campus Research Director	DR. RENATO C. SAGAYNO	

Appendix R: Capstone Project List of Modules

University of Cebu
College of Computer Studies

LIST OF MODULES

Programmer	Modules	Student	Faculty	Admin
Durano, Arcana, Ocliasa, Real	User Management Module			
	1. Create User Account			*
	2. Edit User Account			*
	3. Assign User Roles			*
	4. Reset Password		*	*
	5. Deactivate Account			*
No. of Points (*1 point per module per user*)		0	1	1
Durano, Arcana, Ocliasa, Real	Facial Registration & Recognition Module			
	1. Face Enrollment	*		
	2. Capture Face Template	*		
	3. Real-Time Face Detection	*		
	4. Liveness Detection (Anti-Spoofing)	*		
	5. Log Recognition Event	*		
No. of Points (*1 point per module per user*)		1	0	0
Durano, Arcana, Ocliasa, Real	Attendance Tracking Module			
	1. Auto Timestamp Attendance	*		
	2. Log Attendance to Database	*		
	3. View Attendance	*	*	*

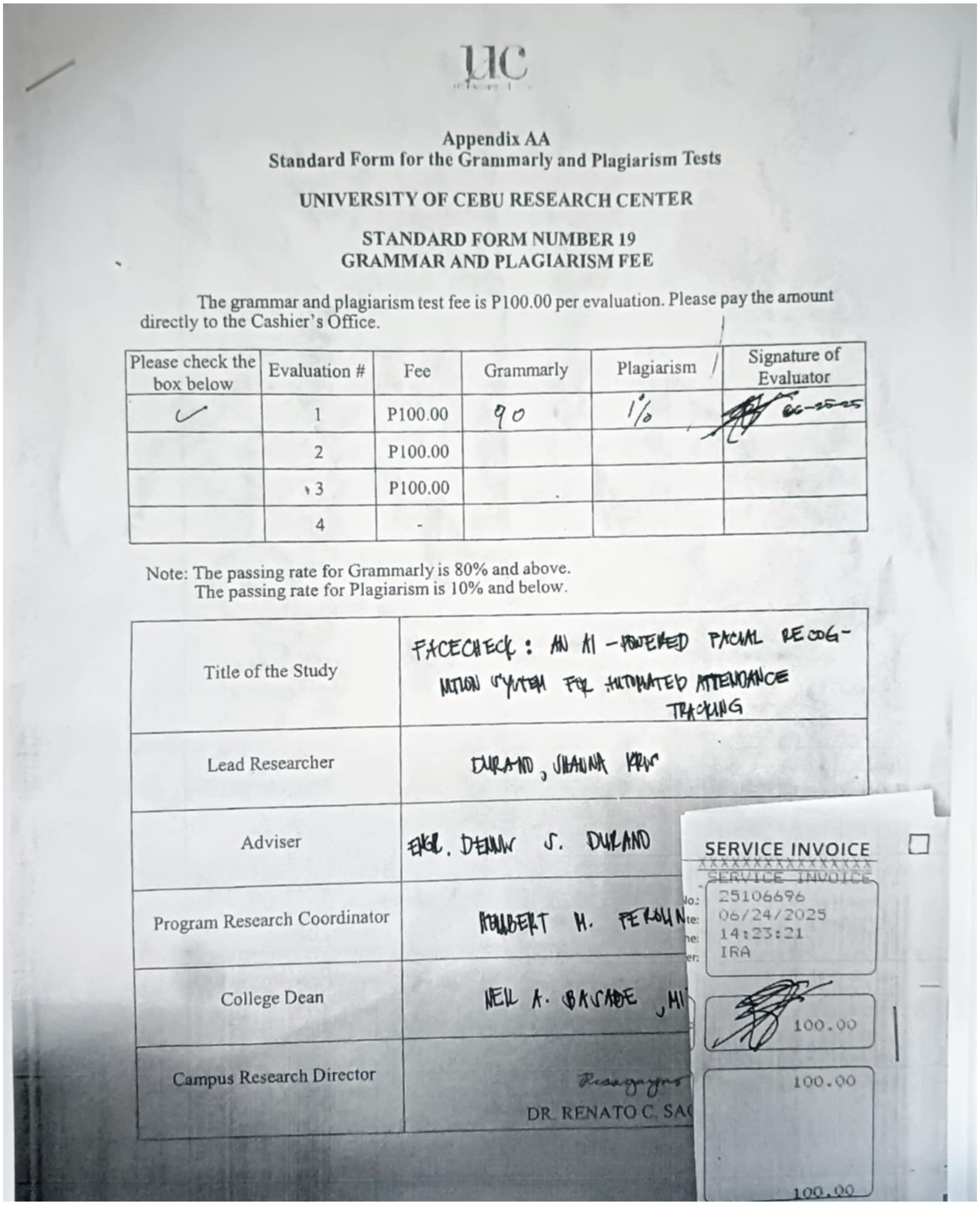
	Record			
	4. Verify Identity on Recognition	*		
	5. Prevent Duplicate Logging	*		
No. of Points (*1 point per module per user*)		1	1	1
Durano, Arcana, Ocliasa, Real	Attendance Monitoring Module			
	1. View Daily Logs	*	*	*
	2. Generate Attendance Reports		*	*
	3. Filter by Class/Date		*	*
	4. Edit/Override Attendance		*	
	5. Export Attendance File (PDF/Excel)		*	*
No. of Points (*1 point per module per user*)		1	1	1
Durano, Arcana, Ocliasa, Real	Class Management Module			
	1. Create Class Schedule			*
	2. Assign Faculty to Classes			*
	3. Enroll Students			*
	4. View Class Lists		*	*
	5. Update Class Info			*
No. of Points (*1 point per module per user*)		0	1	1
Durano, Arcana, Ocliasa, Real	Reports & Analytics Module			
	1. Create Class Schedule			*
	2. Absence Patterns Analytics		*	*
	3. Monthly Attendance Graphs		*	*
	4. Export Report CSV		*	*
	5. Compare Attendance			*

	Across Classes			
No. of Points (*1 point per module per user*)		0	1	1
Durano, Arcana, Ocliasa, Real	System Configuration Module			5
	1. Set Attendance Rules			*
	2. Configure Email Notifications			*
	3. Adjust Recognition Sensitivity			*
	4. Change System Settings			*
	5. Manage Database Settings			*
No. of Points (*1 point per module per user*)		0	0	1
Total Number of Modules		3	5	6

Noted by:

Engr. Dennis Durano
Adviser

Grammar and Plagiarism Form Result



UIC

Appendix AA
Standard Form for the Grammarly and Plagiarism Tests

UNIVERSITY OF CEBU RESEARCH CENTER

STANDARD FORM NUMBER 19
GRAMMAR AND PLAGIARISM FEE

The grammar and plagiarism test fee is P100.00 per evaluation. Please pay the amount directly to the Cashier's Office.

Please check the box below	Evaluation #	Fee	Grammarly	Plagiarism	Signature of Evaluator
✓	1	P100.00	90	1%	06-2025
	2	P100.00			
	3	P100.00			
	4	-			

Note: The passing rate for Grammarly is 80% and above.
The passing rate for Plagiarism is 10% and below.

Title of the Study	FACECHECK: AN AI-POWERED FACIAL RECOGNITION SYSTEM FOR AUTOMATED ATTENDANCE TRACKING
Lead Researcher	DURANO, SHAUNA KRIS
Adviser	ENGR. DENNIS S. DURANO
Program Research Coordinator	ROLBERT H. PEROLIN
College Dean	NEIL A. BASABE, MI
Campus Research Director	DR. RENATO C. SA

SERVICE INVOICE
~~SERVICE INVOICE~~
No: 25106696
Date: 06/24/2025
Time: 14:23:21
Cashier: IRA
100.00
100.00
100.00

BSIT-DURANO 06/25/2025

by UCRC GRAMMAR

General metrics

54,727	7,368	416	29 min 28 sec	56 min 40 sec
characters	words	sentences	reading time	speaking time

Score

90

This text scores better than 90% of all texts checked by Grammarly

Writing Issues

186	12	174
Issues left	Critical	Advanced

Plagiarism

1 %

75 sources

1% of your text matches 75 sources on the web or in archives of academic publications

Capstone Project Minute of Proceeding

UNIVERSITY OF CEBU – MAIN CAMPUS

College of Computer Studies

Capstone Project Minutes of the Proceedings

Date: June 18, 2025 Time: 3:30 PM - 4:30 PM [/] PROPOSAL HEARING

Term: [ ] 1st [ ] 2nd [✓] Summer School Year: SUMMER 2025 [ ] ORAL DEFENSE

Research Title: FaceCheck: An AI-Powered Facial Recognition System for Automated Attendance Tracking

Project Manager: Durano, Jhanna Kris
UI/Ux Designer: Arcana, Sean Joseph
Database Designer: Ocliasa, Niño Rollane
Project Developer: Real, Rovic Steve

Panel Members:
1. Mr. Wilson Gayo
2. Mr. Heubert Ferolino

PARTICULARS:	SUGGESTED BY:	REMARKS
Manuscript - Page numbering is wrong. **Chapter 1** - Follow the "Guidelines in Writing the Objectives of the Study" - Arrange the definition of terms alphabetically **Chapter 3** - Revise the Gantt chart	Basabe	OK
Chapter 2 and 3 Consider revising the following: 1. Objectives of the study (Arrange in chronological order 2. BMC 3. UCD (Possible changes in ACTORS) 4.	Ferolino	OK
Chapter 3 - Include the AI process related	Gayo fw.	ok

Prepared by: **MR. DENNIS DURANO**
Technical Editor/ Censor

Noted by: **MR. NEIL A. BASABE, MIT**
Dean

Transmittal Letter

June 09, 2025

Neil Basabe, MIT
Dean, College of Computer Studies
University of Cebu - Main Campus

Dear Dean Basabe,

We are third-year BSIT students from the University of Cebu – Main Campus, currently undertaking our capstone project, **FaceCheck: An AI-Powered Facial Recognition Attendance Monitoring System**. This study aims to develop a secure, contactless system to enhance student attendance tracking by addressing issues like proxy attendance, manual recording inefficiencies, and data accuracy.

We respectfully seek permission to conduct research within the College of Computer Studies, including distributing survey questionnaires to select students and educators. All data collected will be handled confidentially and used solely for academic purposes, in strict adherence to ethical research standards.

We greatly appreciate your time and consideration and hope for your support in carrying out this study. Thank you very much.

Sincerely,
Jhanna Kris Durano
Project Manager

Noted by:

Engr. Dennis S. Durano
Adviser

Leo Bermudez, MSCS
Research Coordinator

Approved by:

Neil A. Basabe, MIT
CCS Dean

Questionnaire

June 09, 2025

Dear Respondents,

We are a group of 3rd-year Bachelor of Science in Information Technology students from the University of Cebu – Main Campus. We are currently conducting our undergraduate capstone project for IT-CAPSTONE30 (Capstone Project 1) this summer of May and June 2025. Our project is titled **"FaceCheck: An AI-Powered Facial Recognition Attendance Monitoring System."**

This project focuses on developing a secure and contactless attendance monitoring system that uses facial recognition technology to automate the logging of student attendance. The system aims to address common issues such as proxy attendance, time-consuming manual methods, and inaccurate records — while also supporting real-time tracking, privacy protection, and operational reliability even during network interruptions.

To ensure that the system is truly responsive to the needs of its users, we are gathering feedback from both students and educators through this survey. Your honest responses will help us better understand current attendance practices at the University of Cebu, identify common challenges, and evaluate the potential acceptance and usability of the FaceCheck system.

Rest assured that all information collected through this survey will remain **confidential** and used solely for academic and research purposes. Your participation is entirely voluntary, and your input is highly valuable in improving attendance monitoring within our institution.

Thank you very much for your time and support.

Respectfully yours,
Jhanna Kris Durano
Project Manager

Noted by:

Engr. Dennis S. Durano
Adviser

Leo Bermudez, MSCS
Research Coordinator

Neil A. Basabe, MIT
CCS Dean

Survey Questionnaire

FaceCheck — AI-Powered Facial Recognition Attendance System

Instructions: Please answer the following questions honestly. Your responses will help improve attendance monitoring systems at the University of Cebu. All answers will be kept confidential.

Section I: Demographic & Background Information

1. **What is your role in the institution?**
   - ○ Student
   - ○ Educator

2. **What year level or position are you in?**
   (Please select based on your role)
   For Students:
   - ○ 1st Year
   - ○ 2nd Year
   - ○ 3rd Year
   - ○ 4th Year
   - ○ Other (please specify): ___________

   For Faculty/Staff:
   - ○ Junior
   - ○ Senior
   - ○ Managerial
   - ○ Other (please specify): ___________

3. **How often do you attend or monitor attendance in classes?**
   - ○ Daily
   - ○ Several times a week
   - ○ Occasionally
   - ○ Rarely

Section II: Current Attendance Experience

4. **How is attendance currently taken in your classes or department?** (Select all that apply)
   ☐ Manual roll call
   ☐ Paper log sheets
   ☐ Digital system (e.g., online portal)
   ☐ Other (please specify): ______________________________

5. **How satisfied are you with the current attendance process?**
   o Very dissatisfied
   o Dissatisfied
   o Neutral
   o Satisfied
   o Very satisfied

6. **What challenges do you face with the current attendance system?** (Select all that apply)
   ☐ Time-consuming process
   ☐ Inaccurate records due to human error
   ☐ Proxy attendance (someone marking attendance for another)
   ☐ Health risks (e.g., shared materials)
   ☐ Lack of real-time data
   ☐ Other (please specify): ______________________________

7. **In your current attendance system, how much time does monitoring or recording attendance typically take?**
   o Less than 1 minute
   o 1–3 minutes
   o 4–6 minutes
   o More than 6 minutes

8. **Have you ever observed or experienced proxy attendance (someone else signing in for you or another person)?**
   - o Yes, frequently
   - o Yes, occasionally
   - o Yes, rarely
   - o No, never
   - o Prefer not to answer

9. **Have you ever observed or heard of someone manipulating attendance records in your institution?**
   - o Frequently
   - o Occasionally
   - o Rarely
   - o Never

10. **How would you describe the reliability of your current attendance tracking system?**
    - o Very reliable
    - o Somewhat reliable
    - o Unreliable
    - o I'm not sure

Section III: Familiarity and Attitudes Toward Facial Recognition

11. **Have you used or experienced facial recognition technology before?**
    - o Yes
    - o No
    - o Maybe

12. **How comfortable are you with using facial recognition technology for attendance?**
    - o Not comfortable at all
    - o Slightly comfortable

○ Neutral
○ Comfortable
○ Very comfortable

13. **What concerns do you have about using facial recognition for attendance?** (Select all that apply)
☐ Privacy issues
☐ Accuracy of the system
☐ Security of my personal data
☐ System malfunction or errors
☐ None
☐ Other (please specify): ___________

Section IV: Data Privacy and Security Considerations

14. **How important is data privacy to you when it comes to attendance records?**
○ Not important
○ Slightly important
○ Neutral
○ Important
○ Very important

15. **Do you trust AI systems to securely handle your biometric data?**
○ Not at all
○ Slightly
○ Neutral
○ Mostly
○ Completely

16. **Which of the following would make you feel more comfortable using FaceCheck?** (Select all that apply)
☐ Data encryption

- ○ Neutral
- ○ Comfortable
- ○ Very comfortable

13. **What concerns do you have about using facial recognition for attendance?** (Select all that apply)
    - ☐ Privacy issues
    - ☐ Accuracy of the system
    - ☐ Security of my personal data
    - ☐ System malfunction or errors
    - ☐ None
    - ☐ Other (please specify): ___________

Section IV: Data Privacy and Security Considerations

14. **How important is data privacy to you when it comes to attendance records?**
    - ○ Not important
    - ○ Slightly important
    - ○ Neutral
    - ○ Important
    - ○ Very important

15. **Do you trust AI systems to securely handle your biometric data?**
    - ○ Not at all
    - ○ Slightly
    - ○ Neutral
    - ○ Mostly
    - ○ Completely

16. **Which of the following would make you feel more comfortable using FaceCheck?** (Select all that apply)
    - ☐ Data encryption

☐ Limited access to data (only authorized users)
☐ Masking of biometric information
☐ Clear privacy policy
☐ Ability to opt-out
☐ Other (please specify): ___________

Section V: System Usability and Feature Preferences

17. **How important is real-time attendance tracking for you?**
   ○ Not important
   ○ Slightly important
   ○ Neutral
   ○ Important
   ○ Very important

18. **Which features would you find most useful in the attendance system?** (Select all that apply)
   ☐ Real-time attendance dashboard
   ☐ Notifications for absences
   ☐ Ability to correct errors or manual overrides
   ☐ Attendance reports and summaries
   ☐ User-friendly interface
   ☐ Other (please specify): ___________

19. **How easy or difficult do you think it would be to use a facial recognition attendance system?**
   1 - Very difficult
   2 - Difficult
   3 - Neutral
   4 - Easy
   5 - Very easy

20. **Would you prefer an option to manually override the system in case of errors?**
    - ☐ Yes
    - ☐ No

Section VI: Overall Perception and Feedback

21. **Would you support the adoption of FaceCheck at the University of Cebu?**
    - o Yes
    - o No
    - o Maybe

22. **Please provide any additional comments or suggestions:**
    ____
    ____
    ____

Noted by:

Engr. Dennis S. Durano
Adviser